CHAPTER 36
PNEUMATICS

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ILLUSTRATIONS

CHAPTER 36
PNEUMATICS

36-00-00 INTRODUCTION

The pneumatic systems control and distribute bleed air for:

- Pressurization of the cabin and flight compartments
- Conditioned air for interior temperature controls.

GENERAL

Pneumatic Power for the Dash 8 Q400 comes from the bleed-air system. The bleed air system can provide compressed air from one of the following sources.

- Engine(s)
- Auxiliary Power Unit.

It provides compressed air for use by the following aircraft systems:

- Environmental Control System (ECS)
- De-Ice System
- Door Seal Pressurization
- Aircraft Pressurization.

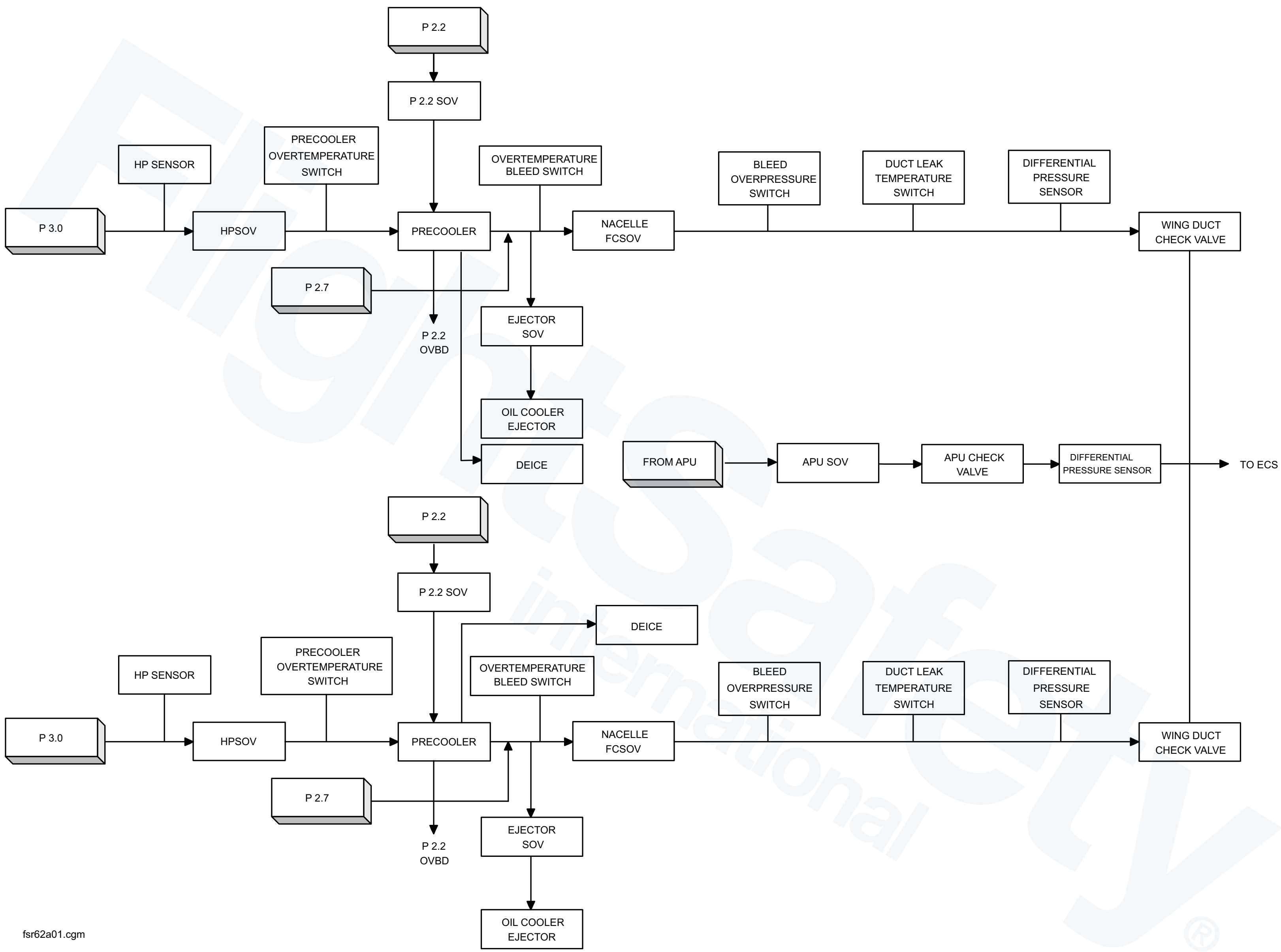

Figure 36-1. Bleed Air System Synoptic Diagram

SYSTEM DESCRIPTION

Refer to:

- Figure 36-1. Bleed Air System Synoptic Diagram.
- Figure 36-2. Air Conditioning and APU Control Panels.

The pneumatic bleed air system can be divided into four subsystems:

- Nacelle (LH and RH)
- Wing (LH and RH)
- Dorsal/Aft fuselage
- APU.

Each subsystem includes ducts, valves, couplings/covers, seals, supports and fasteners.

Each engine has low and high pressure bleed air ports. The bleed air is automatically modulated to supply proper bleed air pressure and flow to service the pneumatic systems. The high pressure bleed supplies adequate pressure and flow capacity at low engine power. The low pressure bleed supplies the flow at all other engine power settings.

Selection of the two bleed control switches on the AIR CONDITIONING control panel turns on the related engine bleed air system. Selection of the bleed air (BL AIR) switchlight on the APU CONTROL panel turns on the APU bleed air.

The Environmental Control System Electronic Control Unit (ECSECU) has left and right digital channels and redundant left and right analog channels. There are protective sensors that monitor bleed air temperature, pressure, flow volume and duct leaks. Either the digital or analog channel is capable of generating a discrete signal to turn on the No.1 or No.2 BLEED HOT caution light on the Caution and Warning Panel (CAWP) if an overtemperature, overpressure or duct leak of the bleed air system occurs.

NOTE

The APU bleed air is able to supply pneumatic pressure when two engine bleed air switches are selected to OFF.

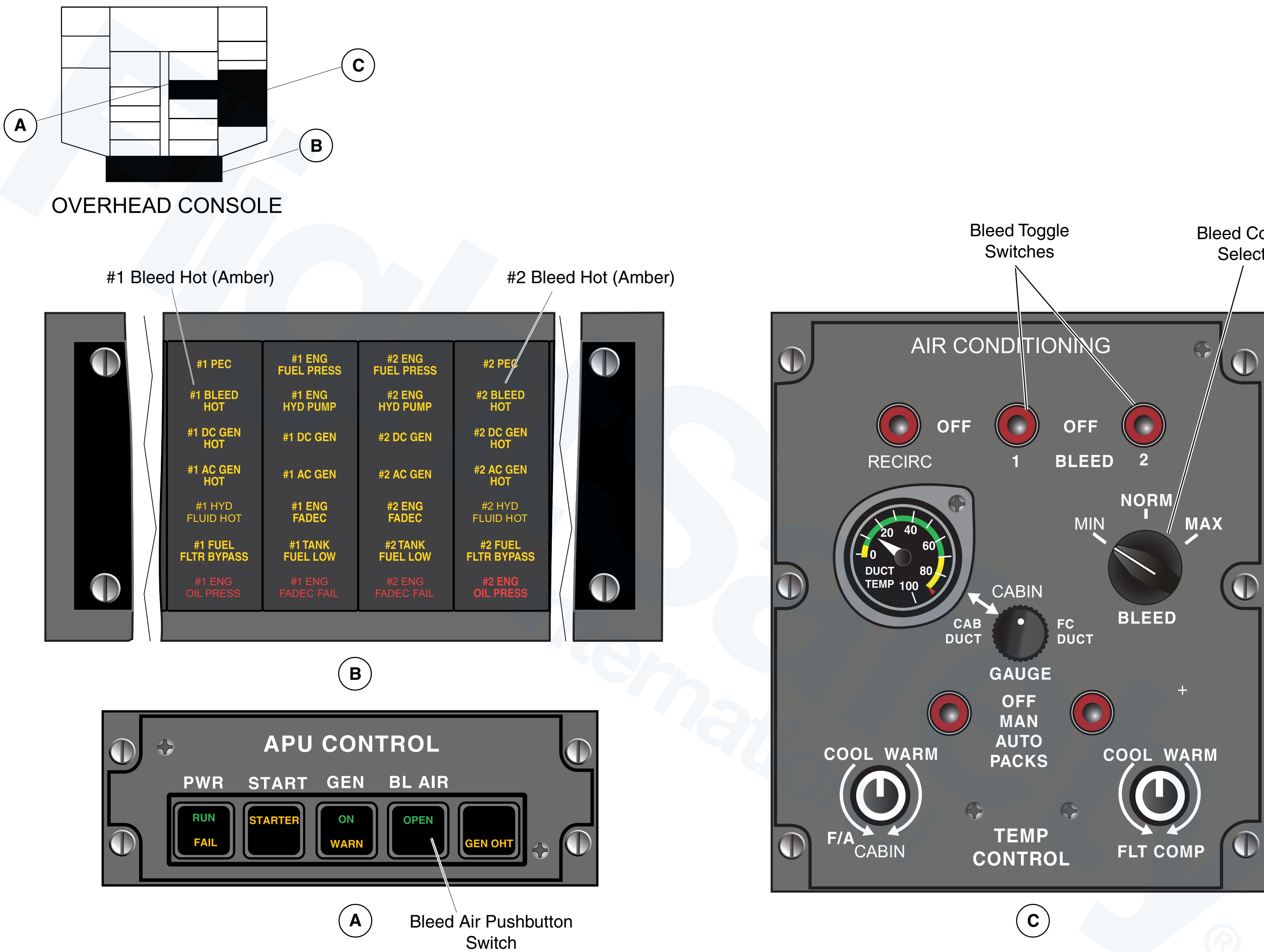

Figure 36-2. Air Conditioning and APU Control Panels

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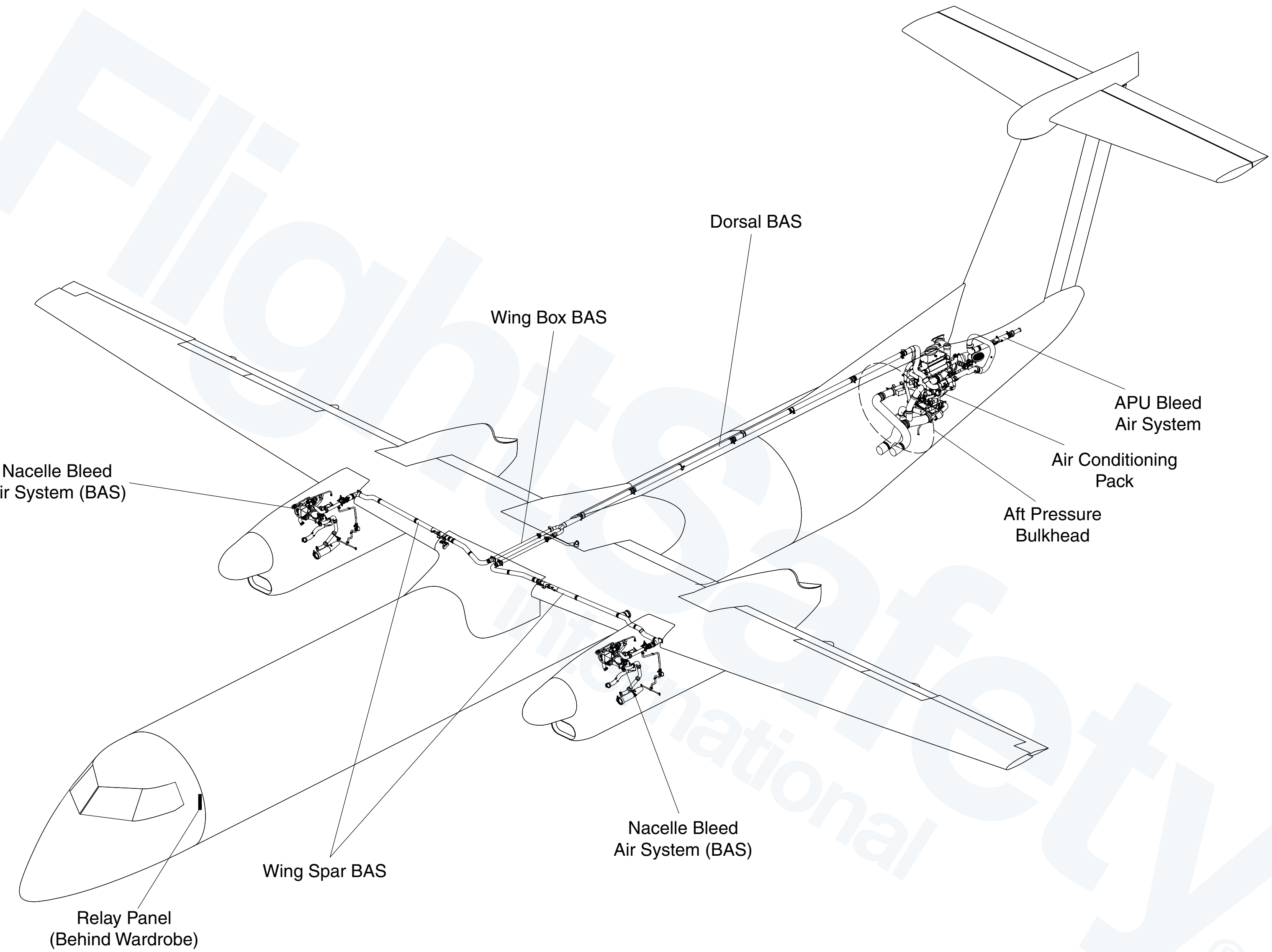


Figure 36-3. Bleed Air System Diagram

36-11-00 BLEED AIR SYSTEM

GENERAL

Refer to Figure 36-3. Bleed Air System Diagram.

The BAS uses air ducts to transfer bleed air from the engines or the APU to the Environmental Control System (ECS).

The bleed air from the engines flows through shrouded ducts in the wing forward spars to the wing center section. Each wing bleed air duct has an over pressure switch, a differential pressure sensor and a check valve to monitor and control the engine bleed air. A sensor in the shroud detects duct leaks.

Shrouded dorsal ducts transfer the air from the wing section to the tail section which joins with the APU duct and supplies the air conditioning packs.

NOTES

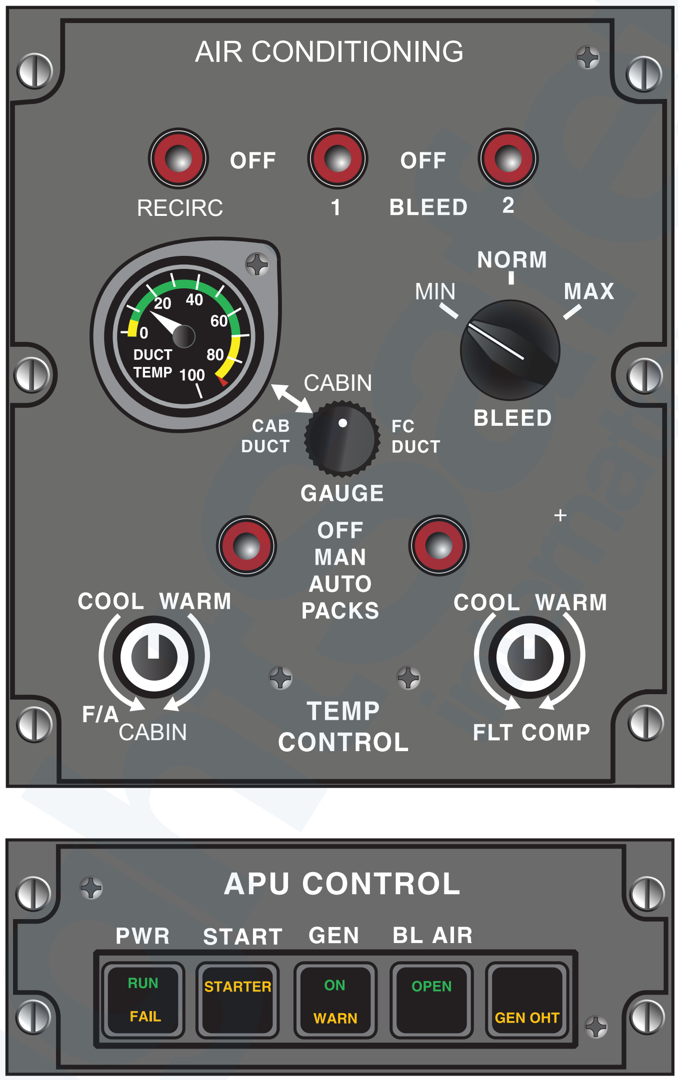


Figure 36-4. Air Conditioning and APU Control Panels

SYSTEM DESCRIPTION

NOTES

Refer to:

- Figure 36-4. Air Conditioning and APU Control Panels.
- Figure 36-5. Bleed Air System.
- Figure 36-6. Bleed Air System Schematic.

Compressed air from the engines or APU is pressure, flow and temperature regulated then supplied to the subsystems.

BLEED switches on the AIR CONDITIONING control panel and the APU control panel determine the bleed air source. The system is protected from overtemperature, overpressurization and leakage.

The left nacelle bleed air system is identical to the right nacelle system.

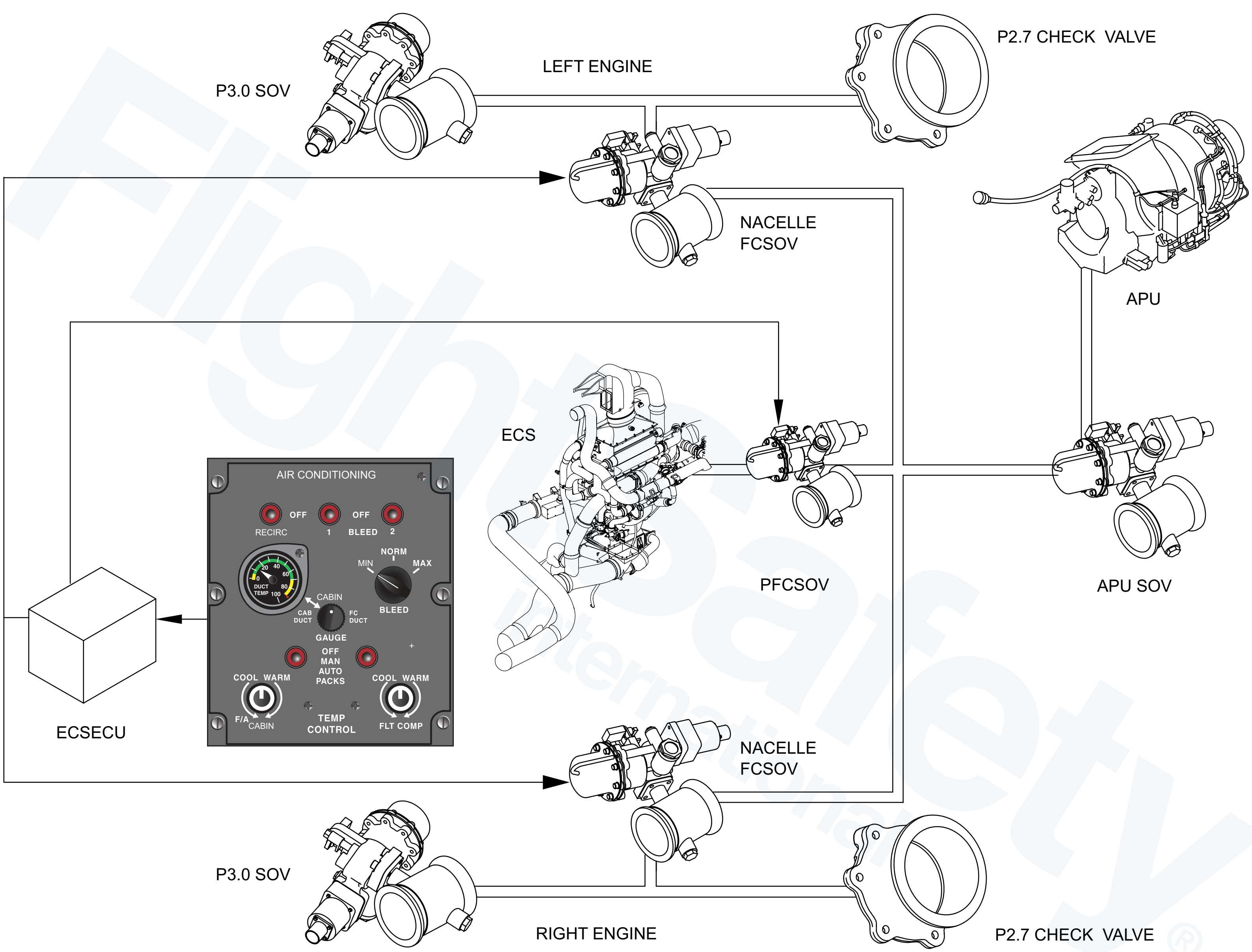


Figure 36-5. Bleed Air System

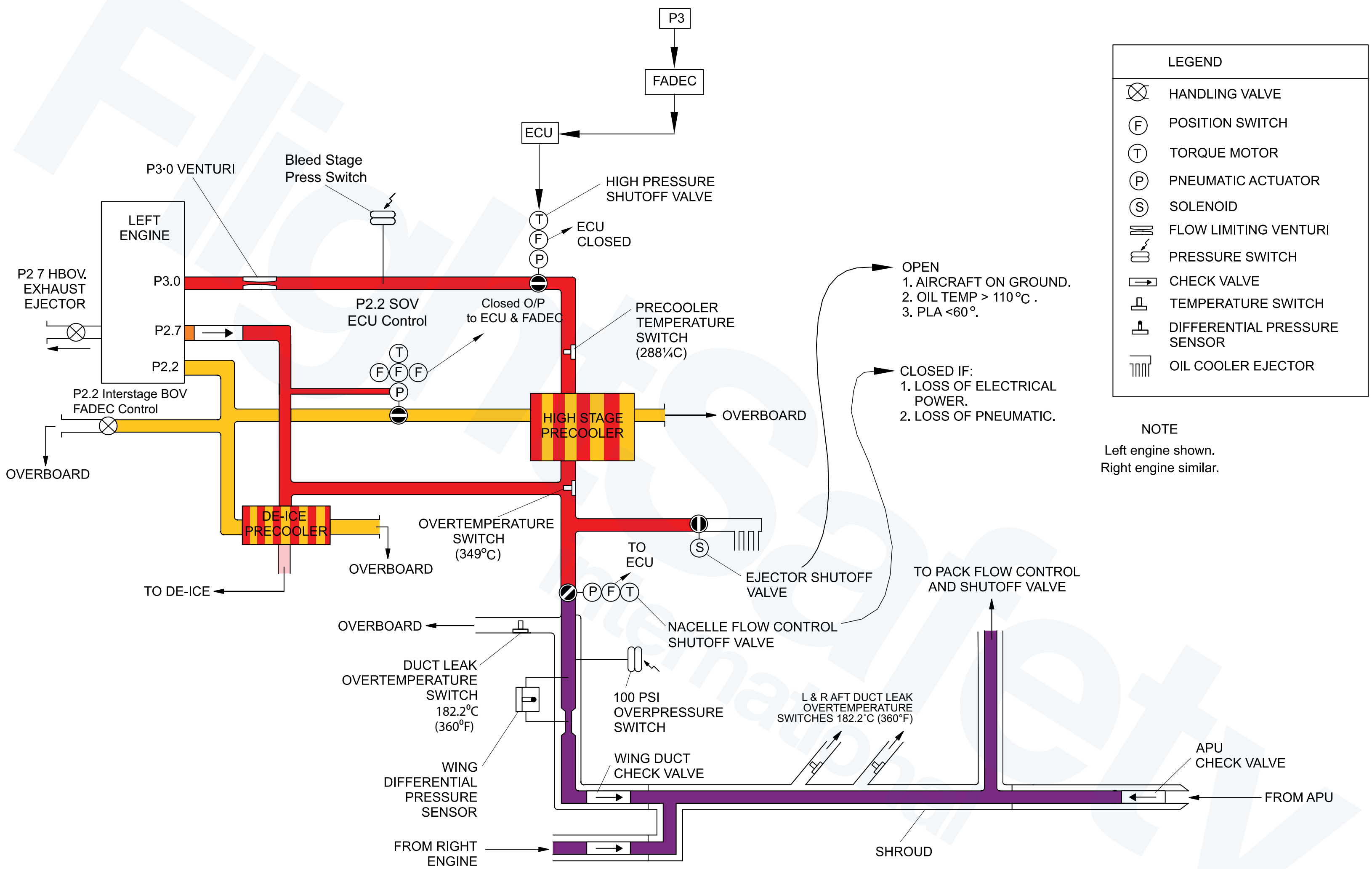
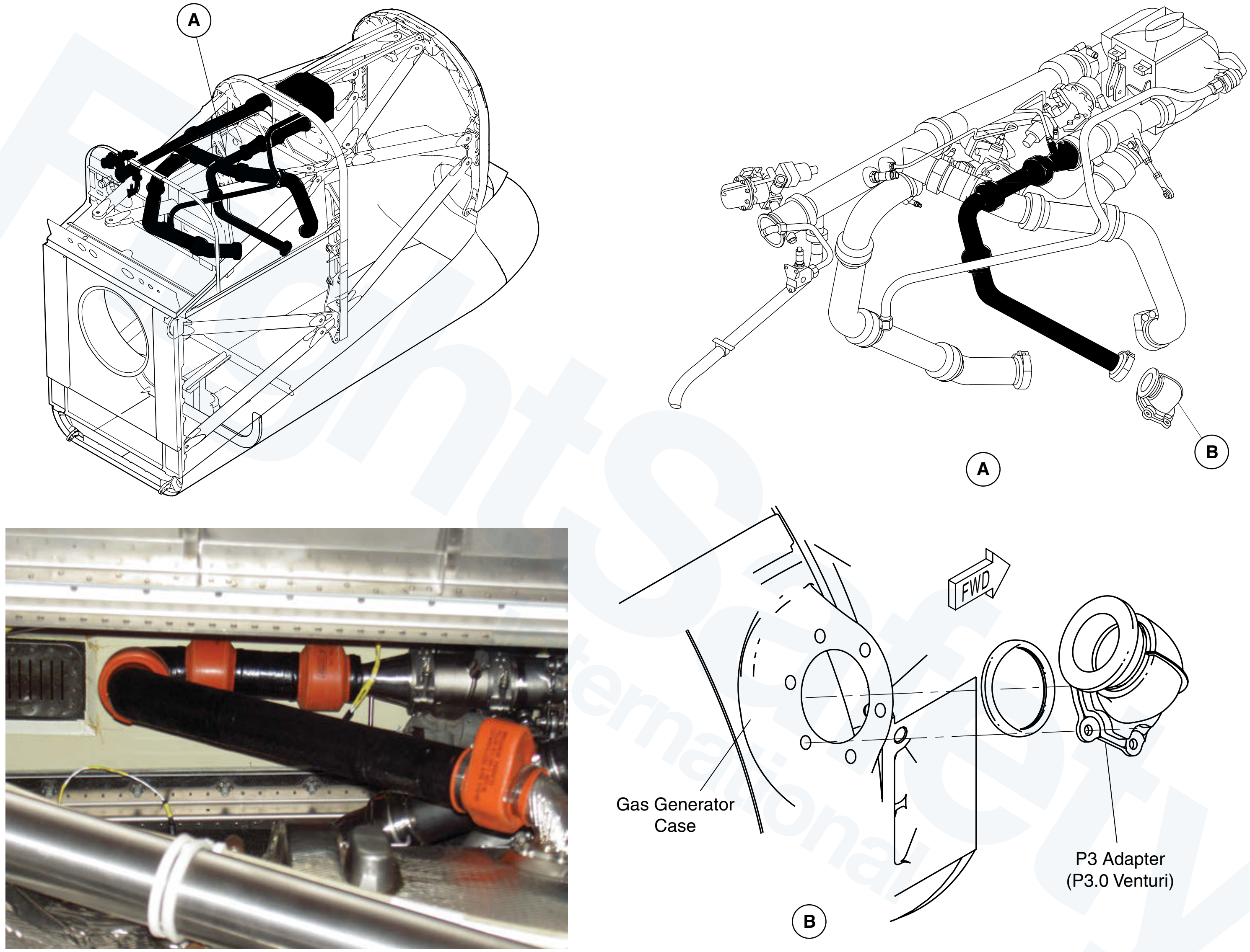


Figure 36-6. Bleed Air System Schematic



A Photo

Figure 36-7. P3.0 Ducts and P3.0 Bleed Air Adapter (P3.0 Venturi)

PRINCIPAL COMPONENTS

P3.0 Ducts

Refer to Figure 36-7. P3.0 Ducts and P3.0 Bleed Air Adapter (P3.0 Venturi).

High pressure ducting, installed between the engine (P3.0) bleed air port and the pre-cooler inlet duct, supplies air flow at low power settings.

The high pressure ducting in the nacelle is enclosed in a composite shroud from the bleed air port to the high pressure shut off valve. This reduces the possibility of ignition in the event of a fuel leak.

P3.0 Venturi

Refer Figure 36-7. P3.0 Ducts and P3.0 Bleed Air Adapter (P3.0 Venturi).

The P3 bleed air adapter (P3.0 Venturi) installed on the engine gas generator case limits the amount of hot, high pressure air entering the bleed air system.

P3.0 Bleed Duct Inspection

Inspect insulation covers on the P3.0 duct sections for signs of dents, holes, tears, and other damage. Also, look for signs of wear, galling, metal pickup, cracks, nicks, burrs, dents, and other signs of damage or wear on the duct.

NOTE

Flex joint covers must have no damage.

Insulation covers that have holes or tears must be replaced.

Damaged insulation covers can be repaired at a repair facility if the hole is less than 2” × 2” or 50mm × 50mm.

NOTES

P2.7 Ducts

Refer to Figure 36-8. P2.7 Bleed Air Duct.

P2.7 ducting, installed between the P2.7 engine bleed air port and the pre-cooler outlet duct, supplies air flow at high power settings.

The P2.7 ducting in the nacelle connects to the high pressure bleed air ducting just upstream of the nacelle shut off valve.

P2.7 Bleed Duct Inspection

When inspecting the P2.7 ducts, look for signs of wear, galling, metal pickup, cracks, nicks, burrs, dents, and other signs of damage or wear.

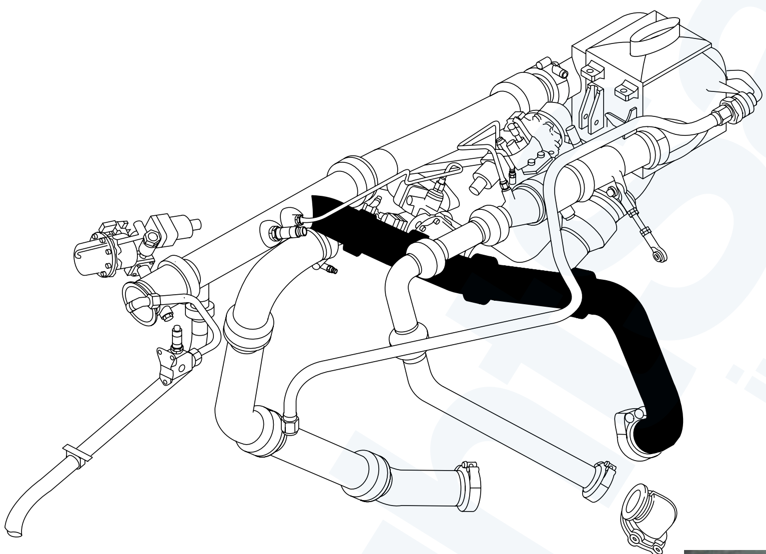
NOTE

Ducts with cracks are not serviceable.

Ducts with only burnish marks are serviceable.

The flange sealing surfaces must have no damage.

Bellow ball joints must have no damage.



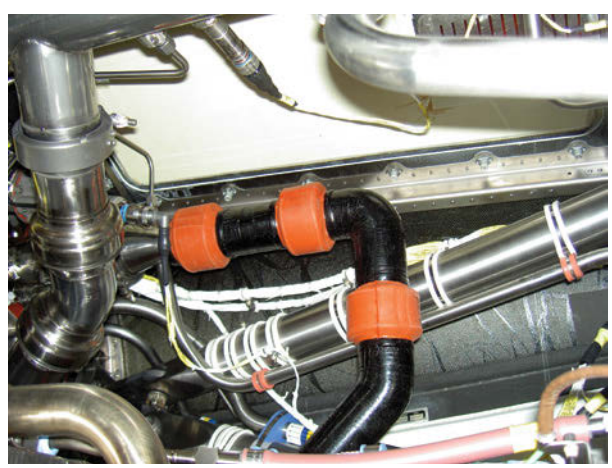


Figure 36-8. P2.7 Bleed Air Duct

Bleed Stage Pressure Switch

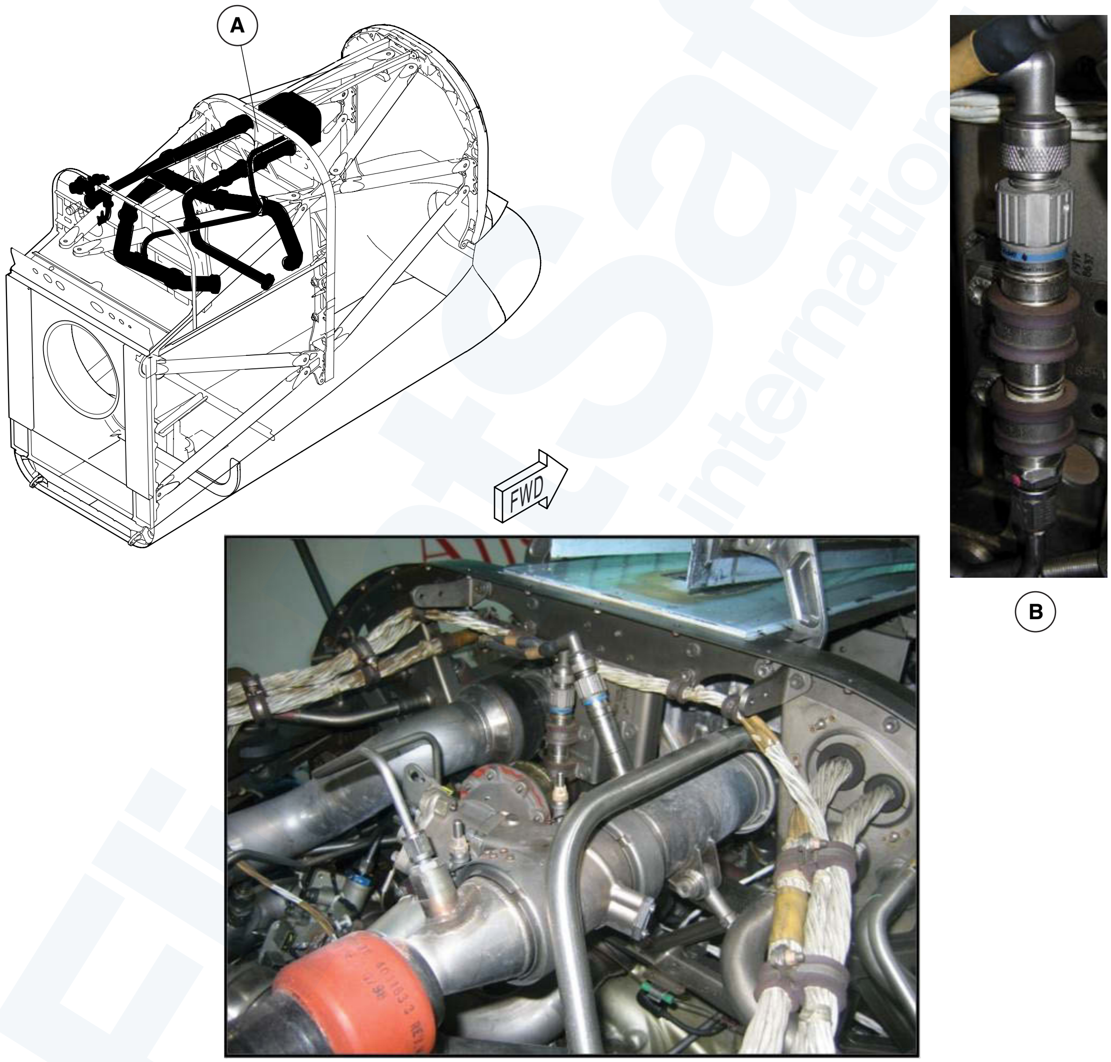
Refer to Figure 36-9. Bleed Stage Pressure Switch.

The bleed stage pressure switch installed on a bracket aft of the pre-cooler senses the high pressure bleed air upstream of the HPSOV.

When the bleed switches are selected off or the deice system is selected on, the HPSOV is controlled by the bleed stage pressure switch.

The switch:

- Opens at 82 psi (565 kPa) increasing pressure
- Closes at 77psi (531 kPa) decreasing pressure.



A BLEED AIR SYSTEM

Figure 36-9. Bleed Stage Pressure Switch

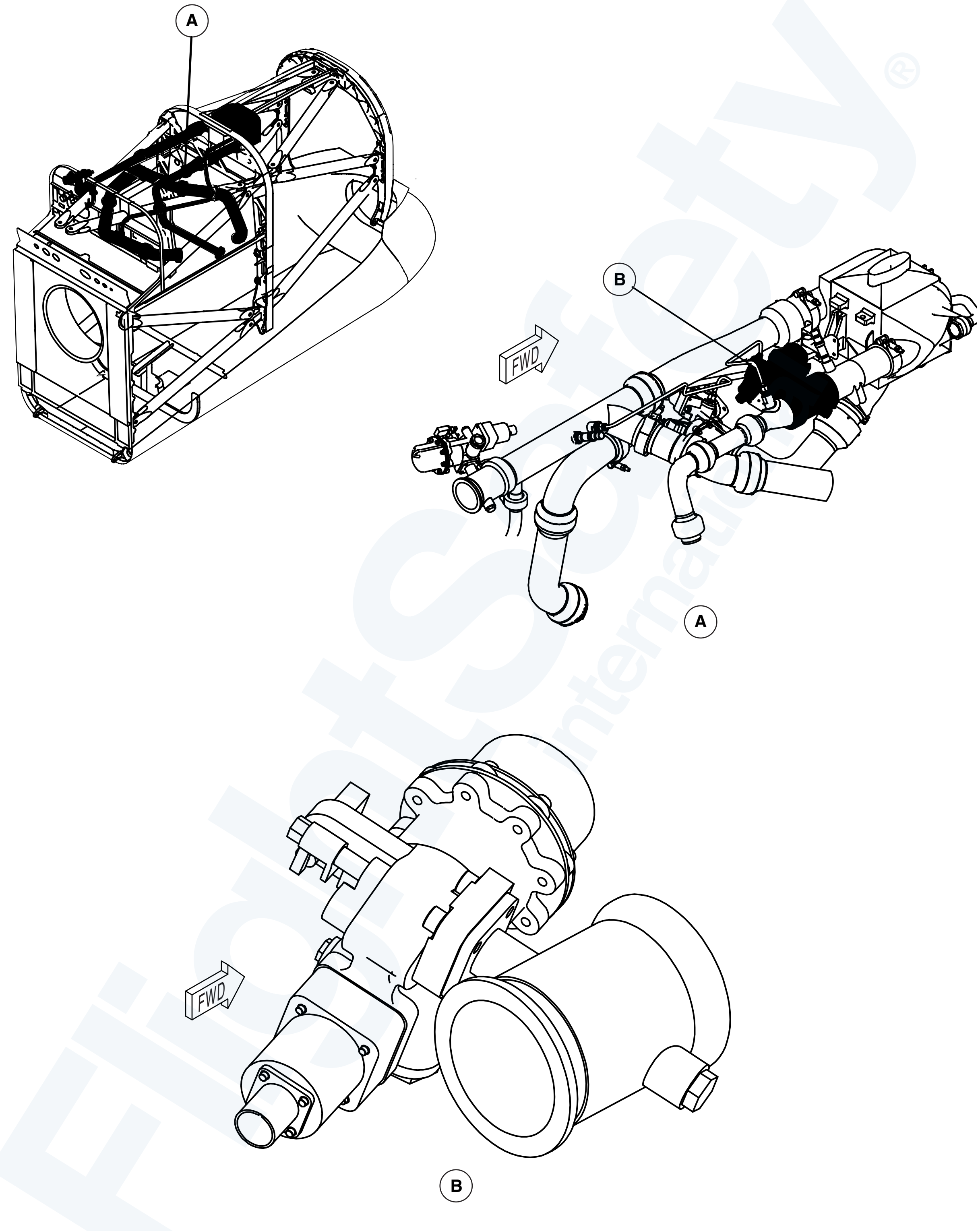


Figure 36-10. High Pressure Shut-Off Valve (HPSOV)

High Pressure Shutoff Valve (HPSOV)

Refer to Figure 36-10. High Pressure Shut-Off Valve (HPSOV).

The HPSOV is a pneumatically operated, torque motor actuated butterfly valve. It has a wire mesh filter upstream of the torque motor to protect the servo from contamination. The valve has a locking pin to lock the valve in the OPEN position and switches that send valve open/closed signals to the ECSECU.

The valve is installed in the high pressure ducting between the engine (P3.0) bleed air port and the pre-cooler.

The HPSOV opens when the torque motor is energized letting P3.0 air supply the bleed air systems.

The ECU uses high pressure (P3.0) information supplied by FADEC or a signal from the bleed stage pressure switch to control the operation of the HPSOV.

Bleed Switch	De-Ice Switch	HPSOV Controlled by:
off	off	Bleed Stage Pressure Switch
off	on	Bleed Stage Pressure Switch
on	on	Bleed Stage Pressure Switch
on	off	ECS ECU

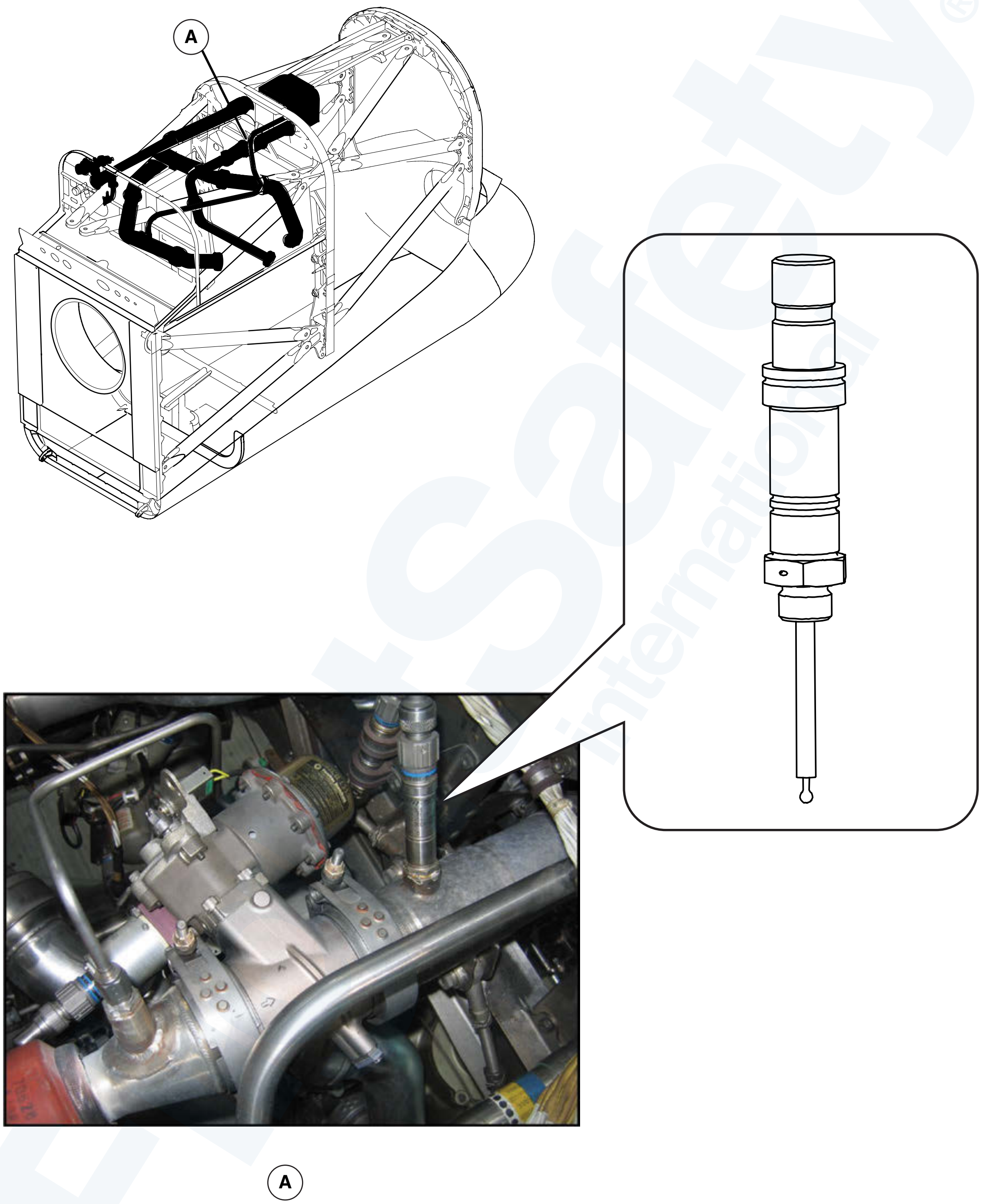


Figure 36-11. Pre-Cooler Overtemperature Switch

Pre-Cooler Overtemperature Switch

Refer to Figure 36-11. Pre-Cooler Overtemperature Switch.

The pre-cooler overtemperature switch installed in the ducting between the HPSOV and the pre-cooler opens if the bleed air temperature exceeds 288°C (550°F) sending a signal to the ECSECU.

The ECU will use this signal to command the P2.2 shut-off valve open, allowing cooling air to pass through the pre-cooler.

NOTES

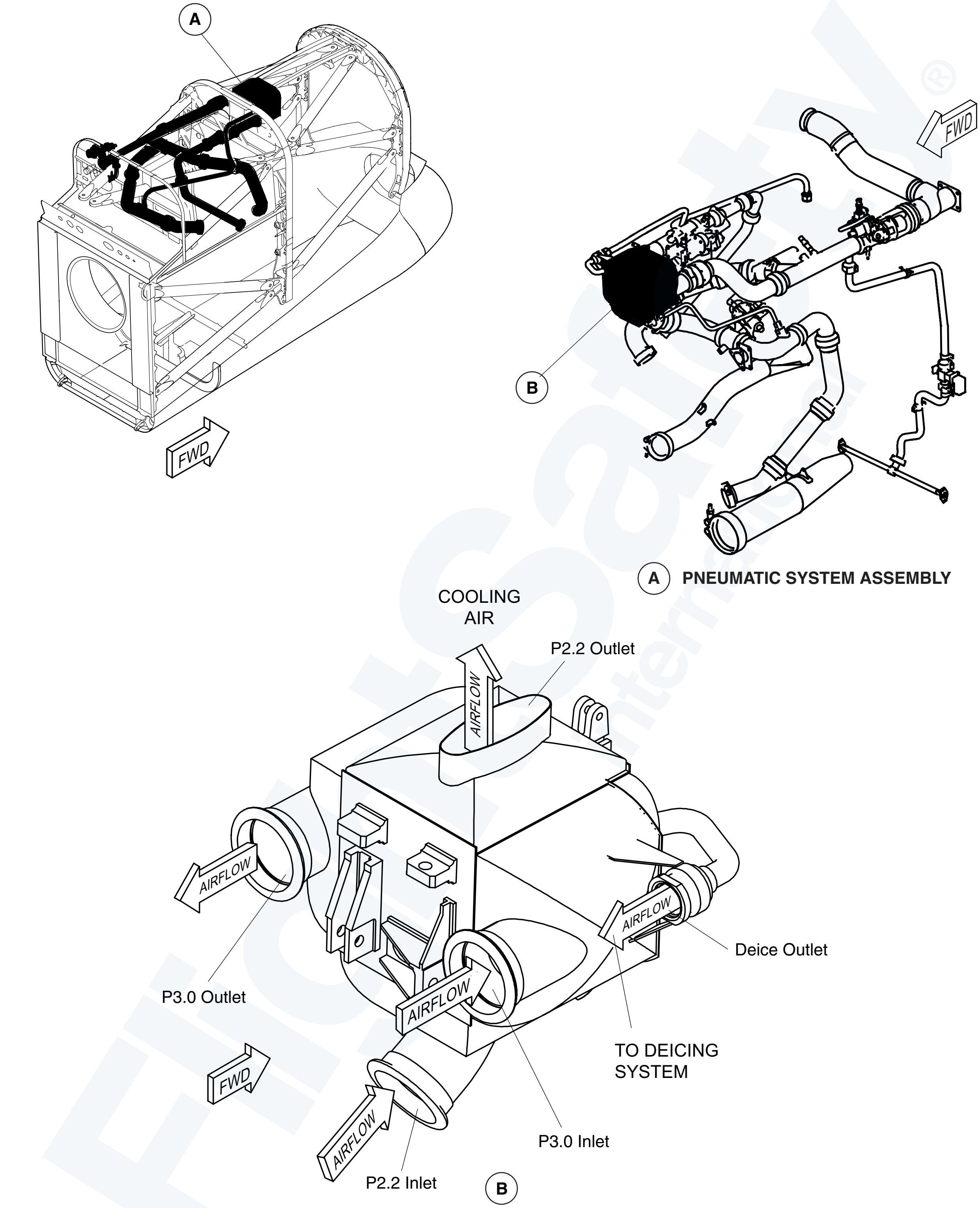


Figure 36-12. Pre-Cooler

Pre-Cooler

Refer to:

- Figure 36-12. Pre-Cooler.
- Figure 36-13. Pre-Cooler.

The pre-cooler is a two part air-to-air heat exchanger mounted to the underside of the spine cowl. The pre-cooler uses low temperature (P2.2) air to cool the high temperature P3.0 bleed air.

A small heat exchanger integral to the pre-cooler reduces the temperature of the bleed air sent to the deice system using a continuous flow of P2.2 bleed air.

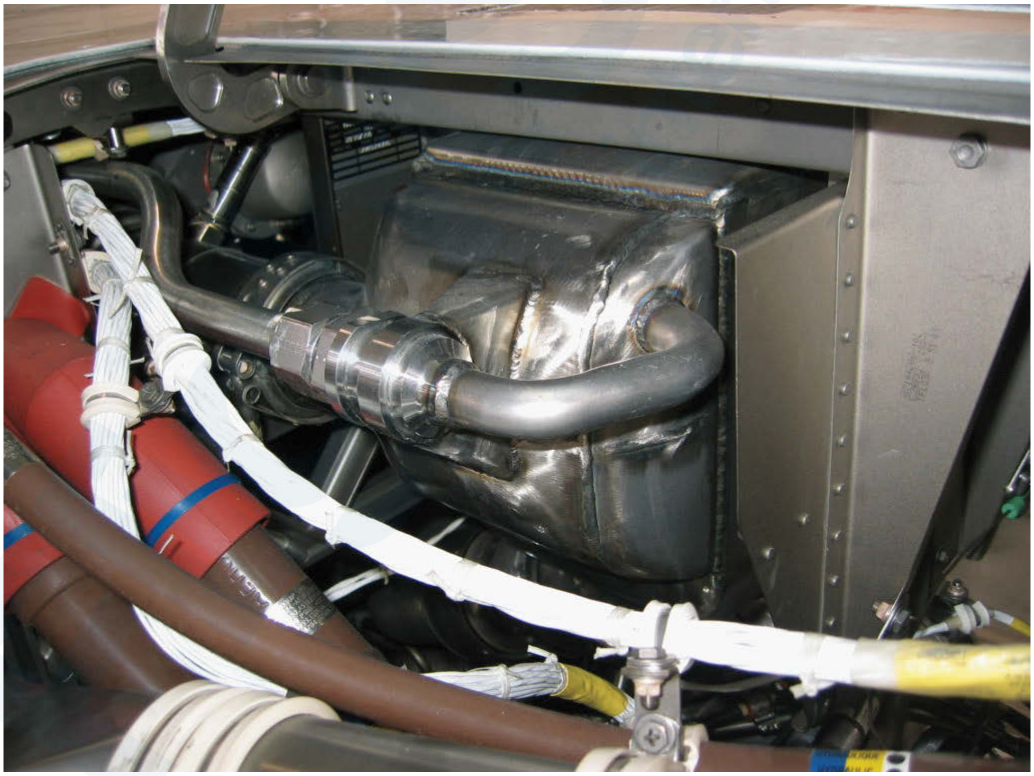


Figure 36-13. Pre-Cooler

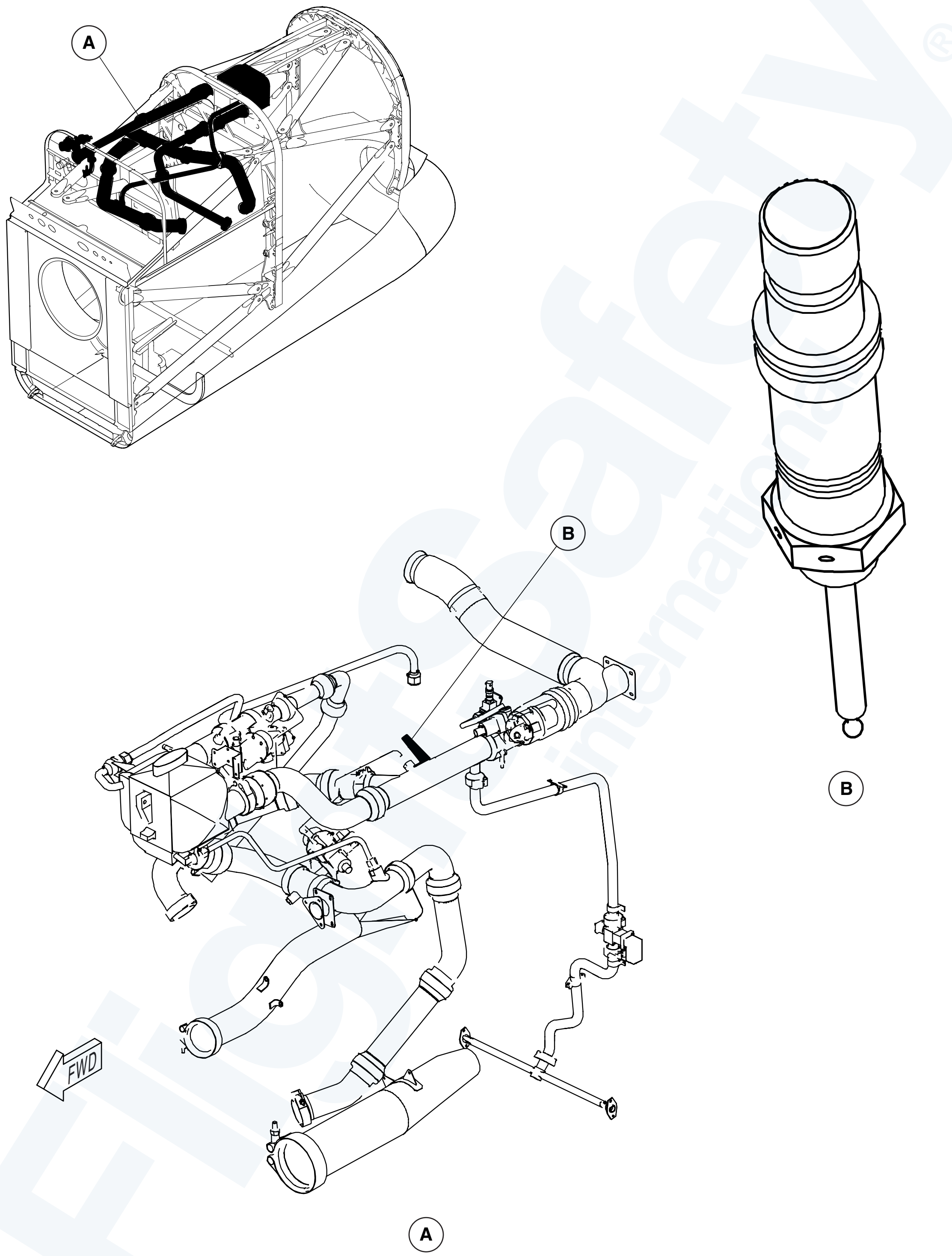


Figure 36-14. Overtemperature Switch

Overtemperature Sensor

Refer to Figure 36-14. Overtemperature Switch.

The overtemperature sensor installed in the ducting upstream of Nacelle Flow Control Shut-Off Valve (NFCSOV) opens if the bleed air temperature exceeds 349°C (660°F) sending a signal to the ECSECU. The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- Nacelle Flow Control Shut-Off Valve (NFCSOV)
- HPSOV
- P2.2 Shut-Off Valve.

The ECU will bring on the related BLEED HOT caution light and log a fault code in the Central Diagnostics System (CDS).

The Resistance Temperature Device (RTD) sensor was introduced as an alternate overtemperature sensor by Bombardier to replace the left and right nacelle overtemperature switches in 2010. Please refer to SB 84-36-03 for details. The new sensors exhibit better life characteristics in extreme operating environments and may reduce nuisance BLEED HOT caution occurrences.

NOTES

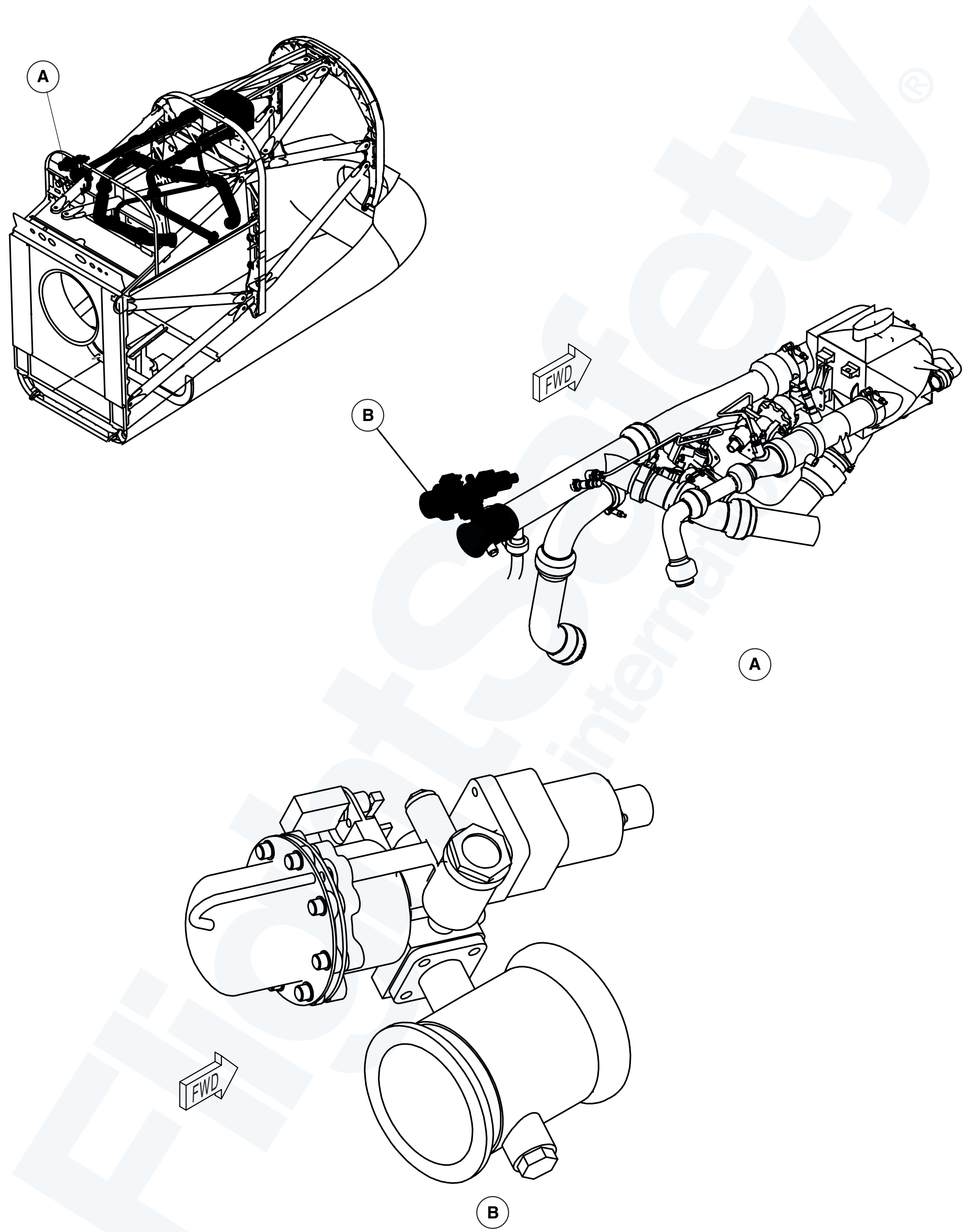


Figure 36-15. Nacelle Flow Control Shut-Off Valve

Nacelle Flow Control Shut-Off Valve (NFCSOV)

Refer to:

- Figure 36-15. Nacelle Flow Control Shut-Off Valve.
- Figure 36-16. Nacelle Flow Control Shut-Off Valve.

The NFCSOV is a pneumatically operated, torque motor actuated butterfly valve, installed between the pre-cooler outlet duct and the wing ducting. It has a wire mesh filter upstream of the torque motor to protect the servo from contamination. The valve has a locking pin to lock the valve in the CLOSED position. A closed position micro-switch sends position signals to the ECSECU (BIT). If electrical power or bleed air is removed, the valve closes.

Selection of the BLEED 1 and/or BLEED 2 switches on the AIR CONDITIONING control panel signals the ECU to open the related NFCSOV. The ECU regulates the NFCSOV according to the settings of the BLEED, PACKS and MIN/NORM/MAX switches.

The ECU digital channels regulate each of the two NFCSOVs to supply 50% of the total flow demand based on differential pressure sensor inputs.

NOTE

There is no AMM procedure to lock out pneumatic valves.

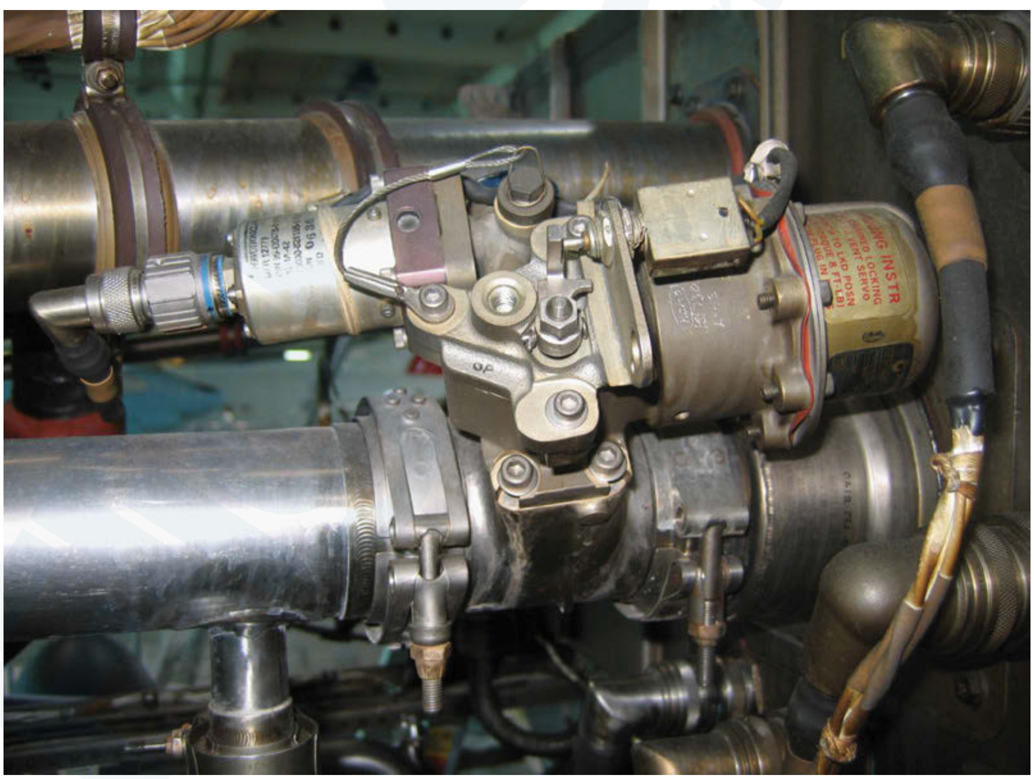


Figure 36-16. Nacelle Flow Control Shut-Off Valve

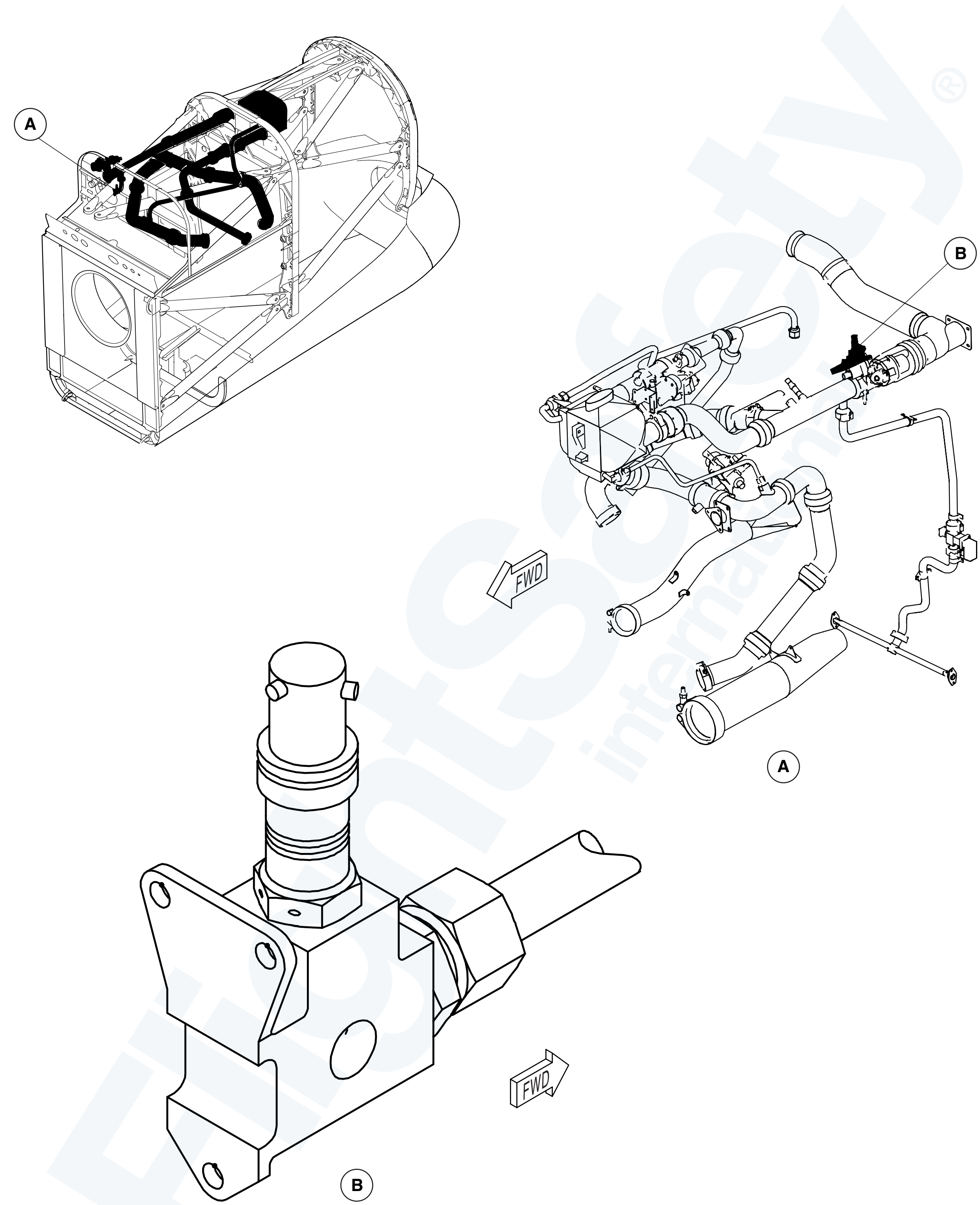


Figure 36-17. Duct Leak Overtemperature Switch

Nacelle Duct Leak Overtemperature Switch

Refer to:

- Figure 36-17. Duct Leak Overtemperature Switch.
- Figure 36-18. Leak Detection Concept Schematic.

The nacelle duct leak overtemperature switch is installed in a housing next to the nacelle shut-off valve, and is connected to the shroud outlet with rigid tubing. The switch opens if the temperature exceeds 182.2°C (360°F) sending a signal to the ECSECU.

The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- NFCSOV
- HPSOV
- P2.2 SOV.

The ECU will bring on the related BLEED HOT caution light and log a fault code in the Central Diagnostics System (CDS).

NOTES

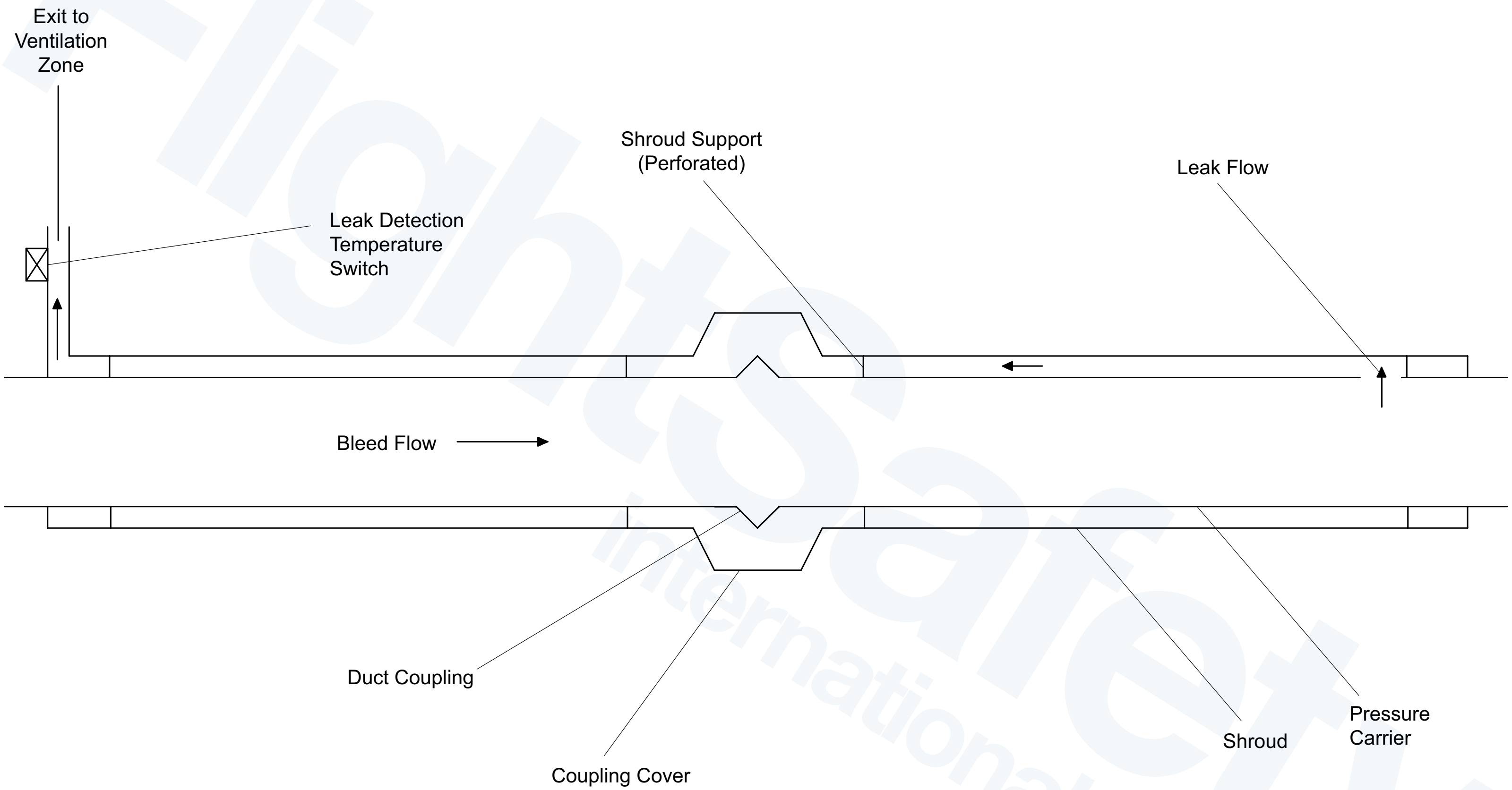


Figure 36-18. Leak Detection Concept Schematic

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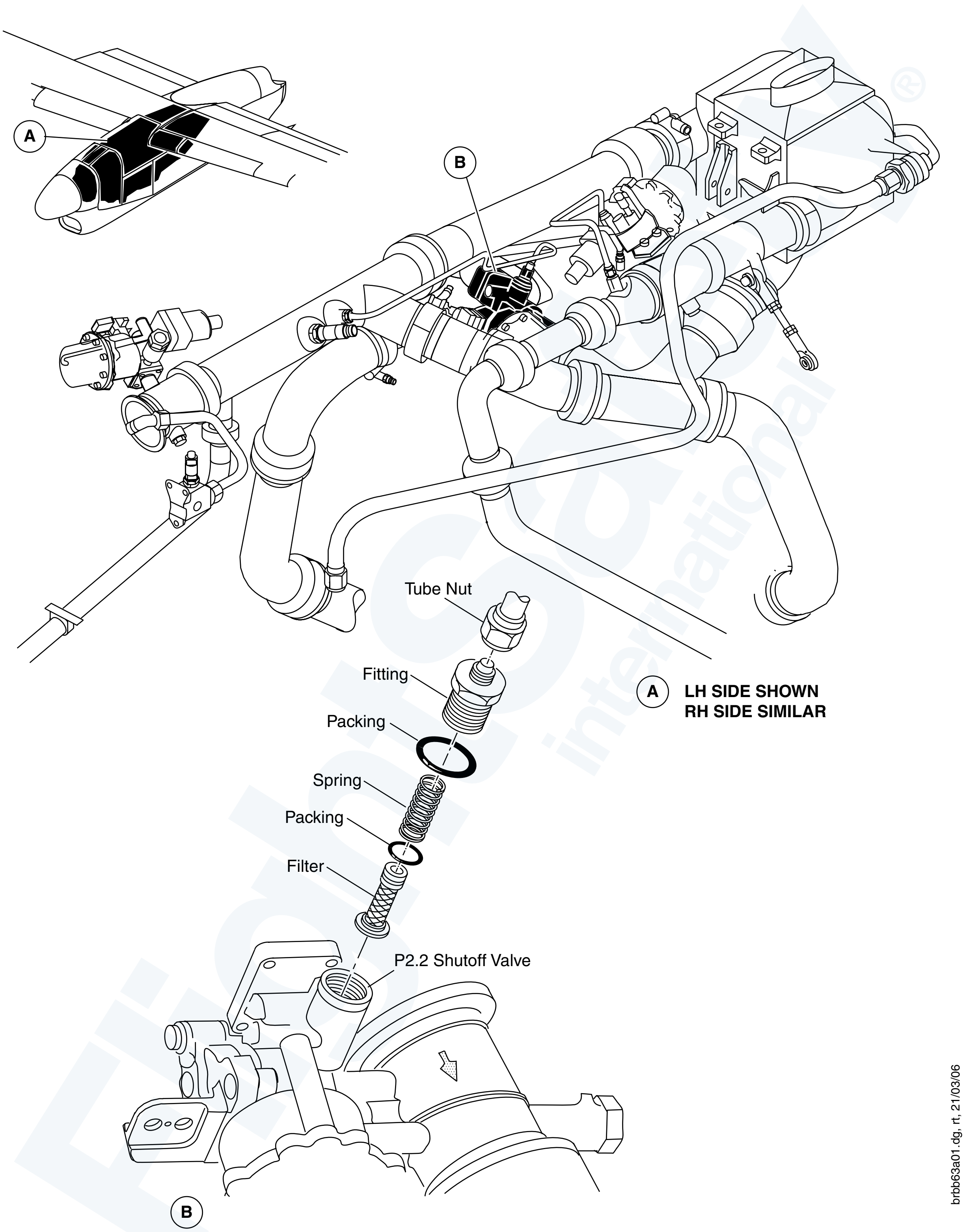

Figure 36-19. P2.2 Shut-off Valve

P2.2 Shut-off Valve

Refer to:

- Figure 36-19. P2.2 Shut-off Valve.
- Figure 36-20. P2.2 Shut-off Valve.

The P2.2 shut-off valve is a pneumatically operated, torque motor controlled butterfly valve installed in-line with the P2.2 low pressure ducting and the pre-cooler. The valve has a wire mesh filter before the torque motor to protect the servo from contamination. The valve has two closed position micro-switches that send valve position information to the ECU and to the Full Authority Digital Electronic Control (FADEC).

The valve is opened by the ECU when it receives a signal from the pre-cooler inlet overtemperature switch. When opened, cooling air is allowed to flow across the pre-cooler.

Figure 36-20. P2.2 Shut-off Valve

Wing Bleed Air Ducts

Refer to Figure 36-21. Wing Bleed Air Ducts.

Titanium wing bleed air ducts are installed on the wing front spar and in the wing box on top of the fuselage in the center section of the wing.

The wing bleed air ducts are an assembly of the following pieces of bleed air ducts:

- Firebox duct
- Inboard firebox duct
- Wing spar duct
- Fairing duct
- Wingbox duct
- Wye duct.

The ducts contain wing differential pressure sensors, overpressure switches and wing duct check valves.

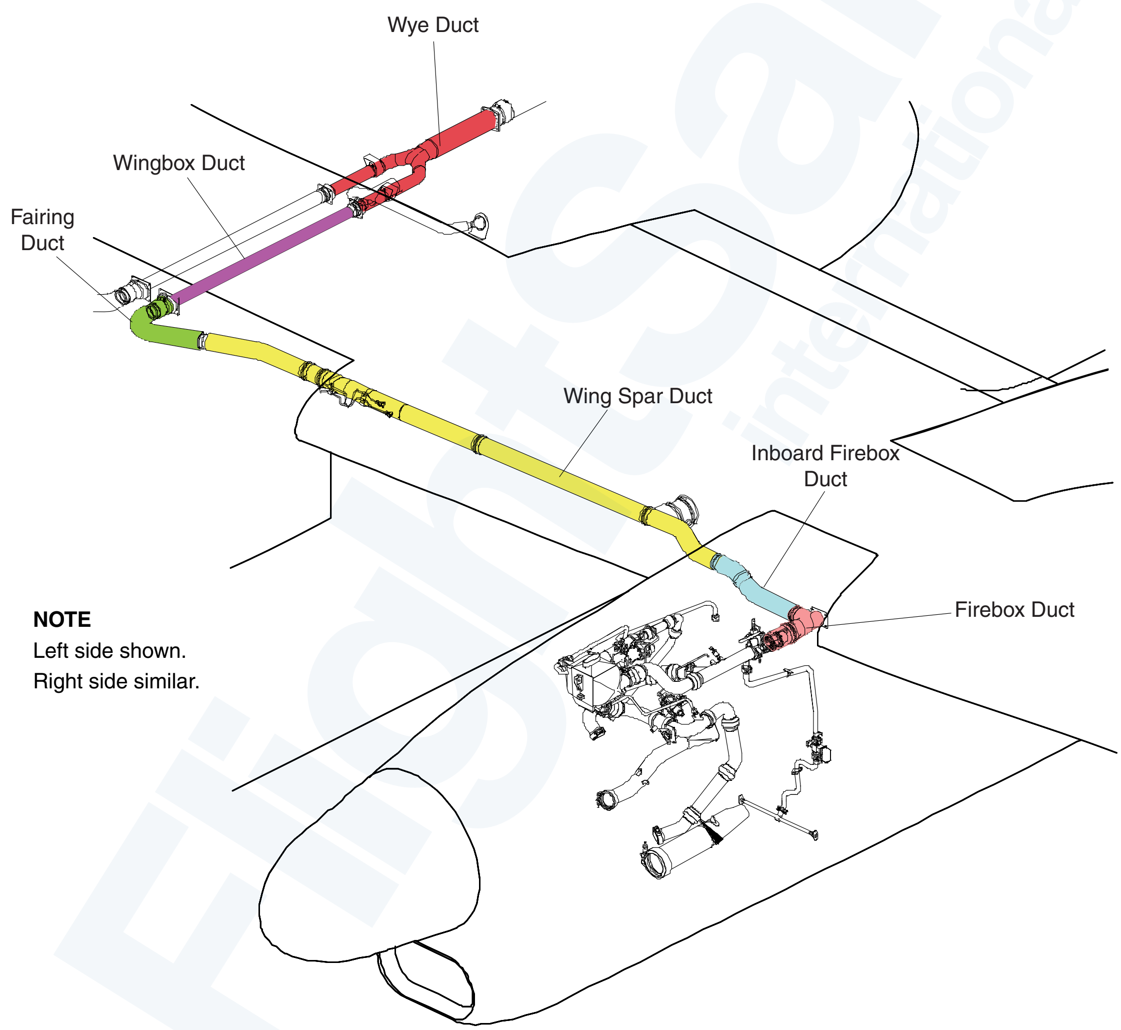

Figure 36-21. Wing Bleed Air Ducts

Overpressure Switch

Refer to Figure 36-22. Overpressure Switch.

The overpressure switch is installed on a bracket in the wing leading edge, and is connected to the bleed air ducting with rigid tubing. The switch closes if the pressure in the duct exceeds 100 psi (690 kPa).

The ECU will use this signal to shut down the related bleed air system by closing the following valves:

- NFCSOV
- HPSOV
- P2.2 SOV.

The ECU will bring on the related BLEED HOT caution light and log a fault code in the Central Diagnostics System (CDS).

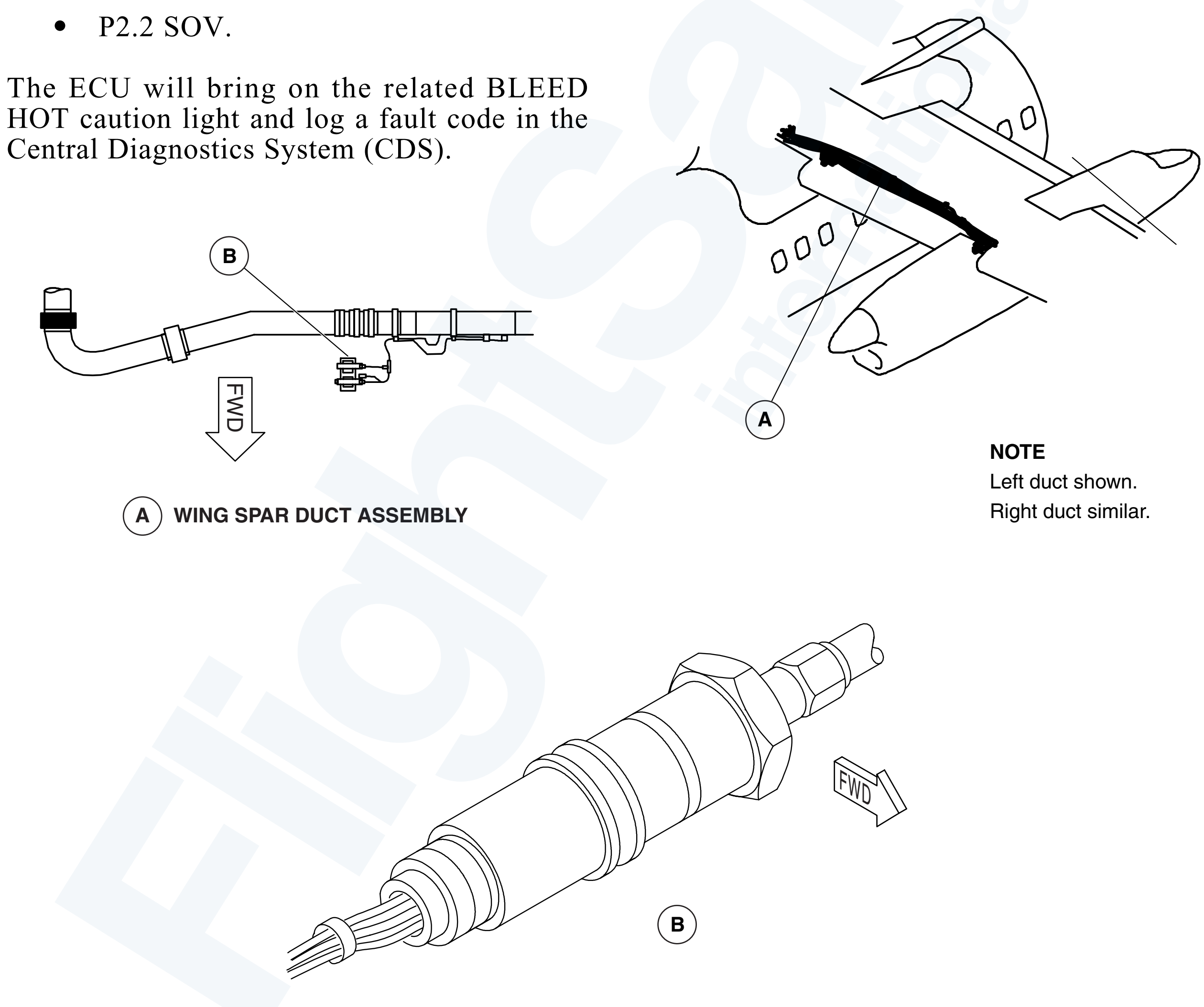

Figure 36-22. Overpressure Switch

Wing Differential Pressure Sensor & Wing Duct Check Valve

Refer to Figure 36-23. Wing Differential Pressure Sensor and Wing Duct Check Valve.

The wing differential pressure sensor is installed on a bracket in the wing root leading edge next to the duct overpressure switch and is connected to bleed air ducting with rigid tubing. The differential pressure sensor has two pressure ports. One port senses the pressure before the wing venturi and the other port senses the pressure at the throat of the venturi. The differential pressure is used to compute the flow through the venturi.

The ECU uses this data together with pressure and temperature data to regulate the bleed air flow through the NFCSOV and to balance the flow of bleed air from the engines.

A dual flapper, spring loaded check valve is installed in the bleed air duct to prevent reverse flow to each nacelle.

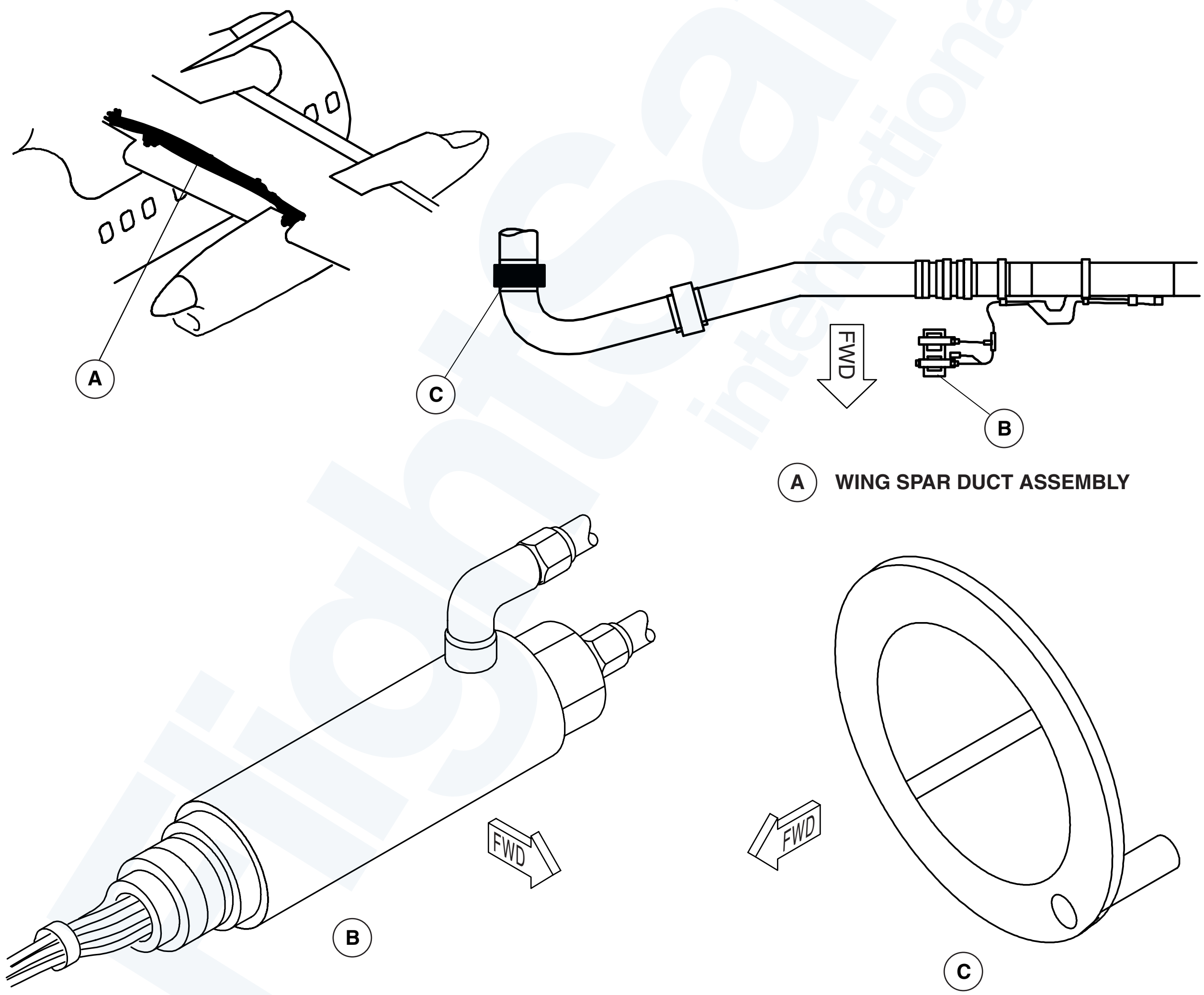

Figure 36-23. Wing Differential Pressure Sensor and Wing Duct Check Valve

Dorsal and Aft Fuselage Bleed Air Ducts

Refer to Figure 36-24. Dorsal and Aft Fuselage Bleed Air Ducts.

Titanium dorsal bleed air ducts, installed on top of the fuselage, run from the wye duct in the wing center section along the top of the fuselage to the dorsal fin at the aft fuselage. The dorsal ducting incorporates a pair of bellows ball joints that add flexibility to the assembly.

The dorsal and aft fuselage bleed air ducts have shrouds and contain a duct leak overtemperature switch and an APU duct leak overtemperature switch.

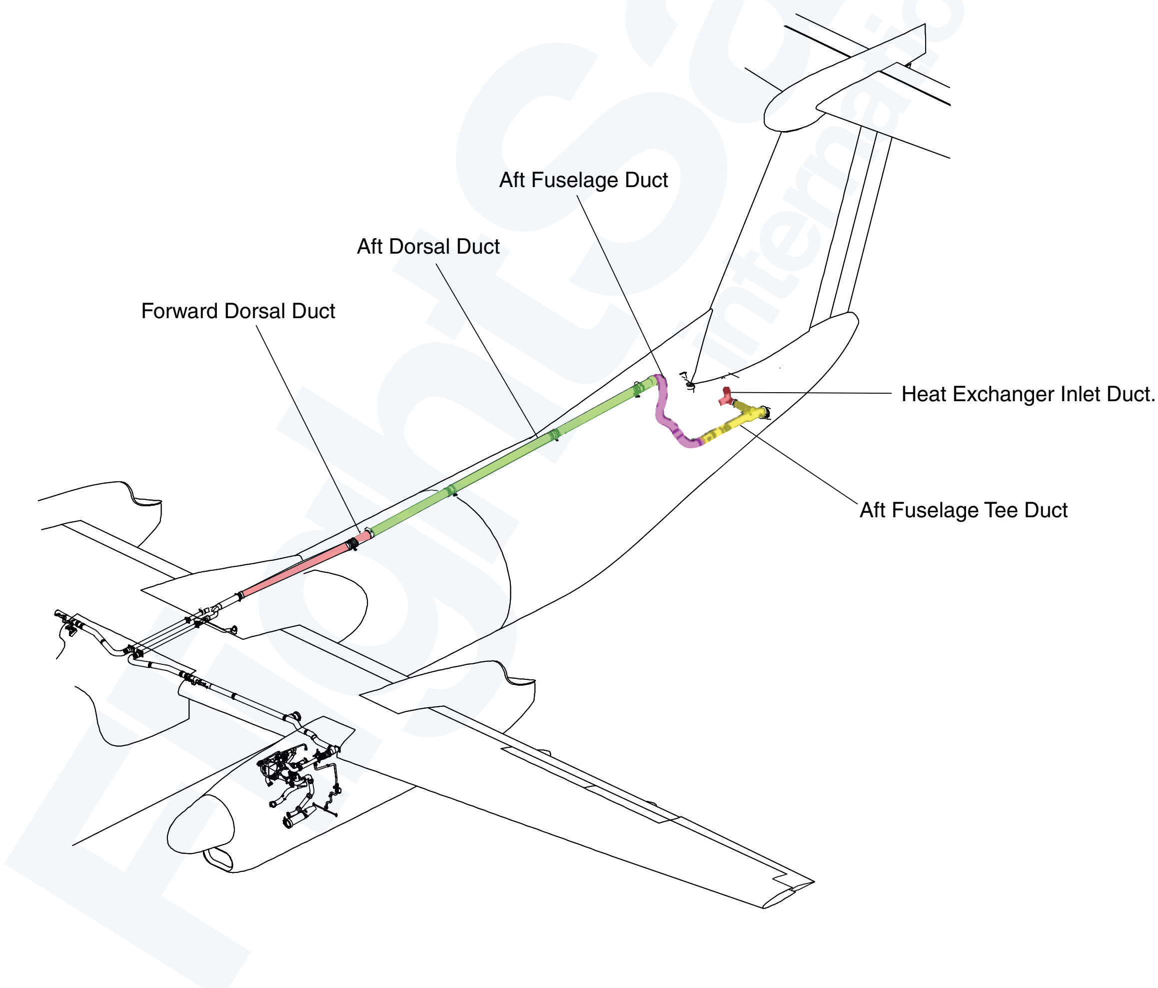

Figure 36-24. Dorsal and Aft Fuselage Bleed Air Ducts

APU Differential Pressure Sensor

Refer to Figure 36-25. Dorsal and Aft Fuselage Bleed Air Ducts.

The APU differential pressure sensor is installed on the APU bleed air ducting in the aft fuselage and is connected by rigid tubing.

The differential pressure sensor has two pressure ports. One port senses the pressure upstream of the APU venturi and the other port senses the pressure at the throat of the venturi. The differential pressure is used to compute the flow through the venturi.

The dorsal and aft fuselage bleed air ducts have shrouds and contain two duct leak overtemperature switches that are installed side by side.

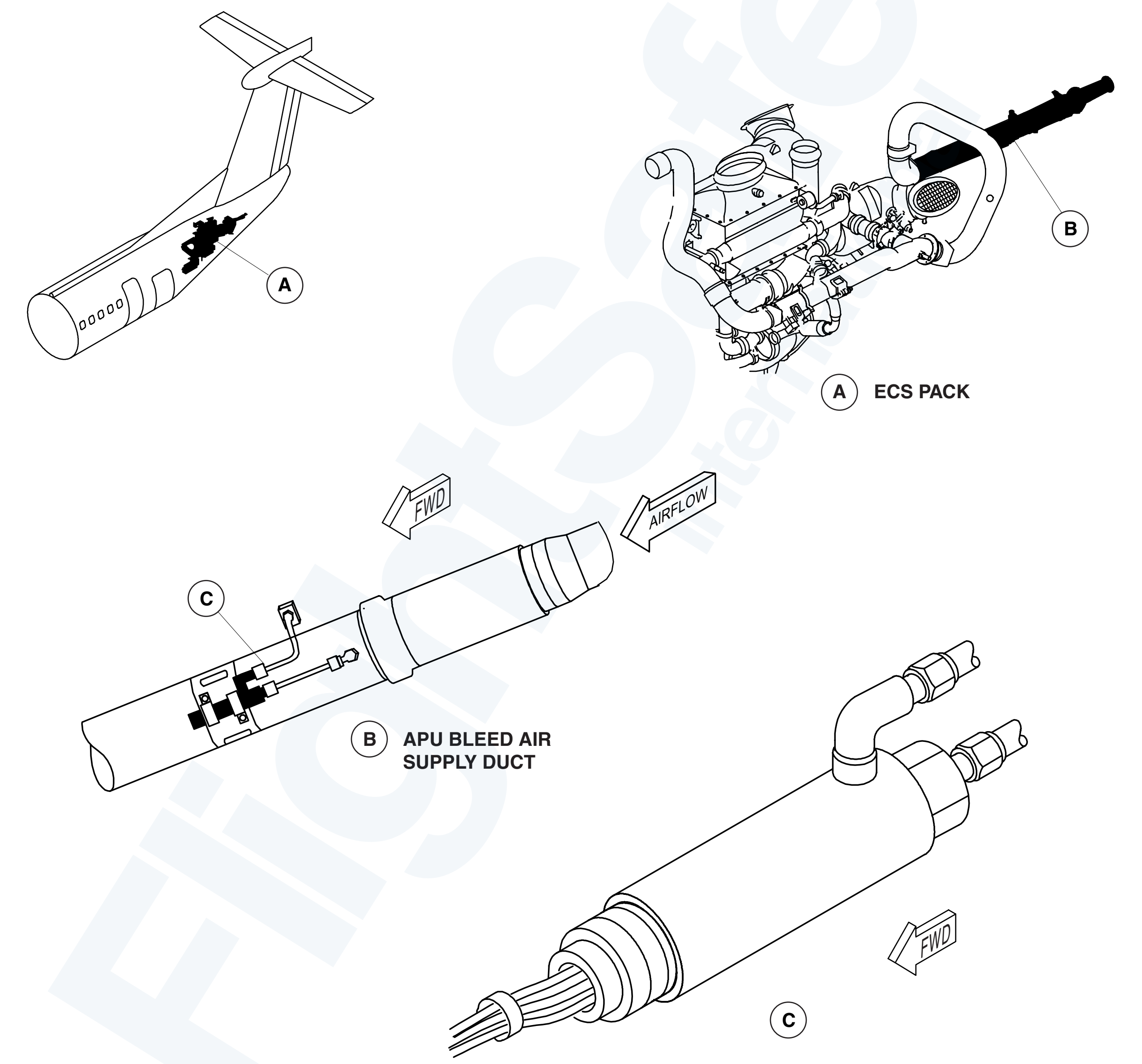

Figure 36-25. Dorsal and Aft Fuselage Bleed Air Ducts

Duct Leak Overtemperature Switches - Aft Fuselage

Refer to Figure 36-26. Aft Fuselage Duct Leak Overtemperature Switch.

Two aft fuselage duct leak overtemperature switches are installed in the same bracket on the aft fuselage T-duct.

The aft fuselage duct leak overtemperature switches detect a bleed air leak in the ducts near the air conditioning pack inlet. If the temperature in the duct is greater than 360°F (182.2°C), the switch contacts open.

If an overtemperature switch detects a leak, the ECU closes the related NFCSOV, HPSOV and P2.2 SOV.

The APU FADEC will close the APU bleed valve. The BL AIR OPEN advisory light on the APU panel goes OFF.

The ECU also illuminates the related BLEED HOT caution light and records a fault code in the CDS to show a duct leak.

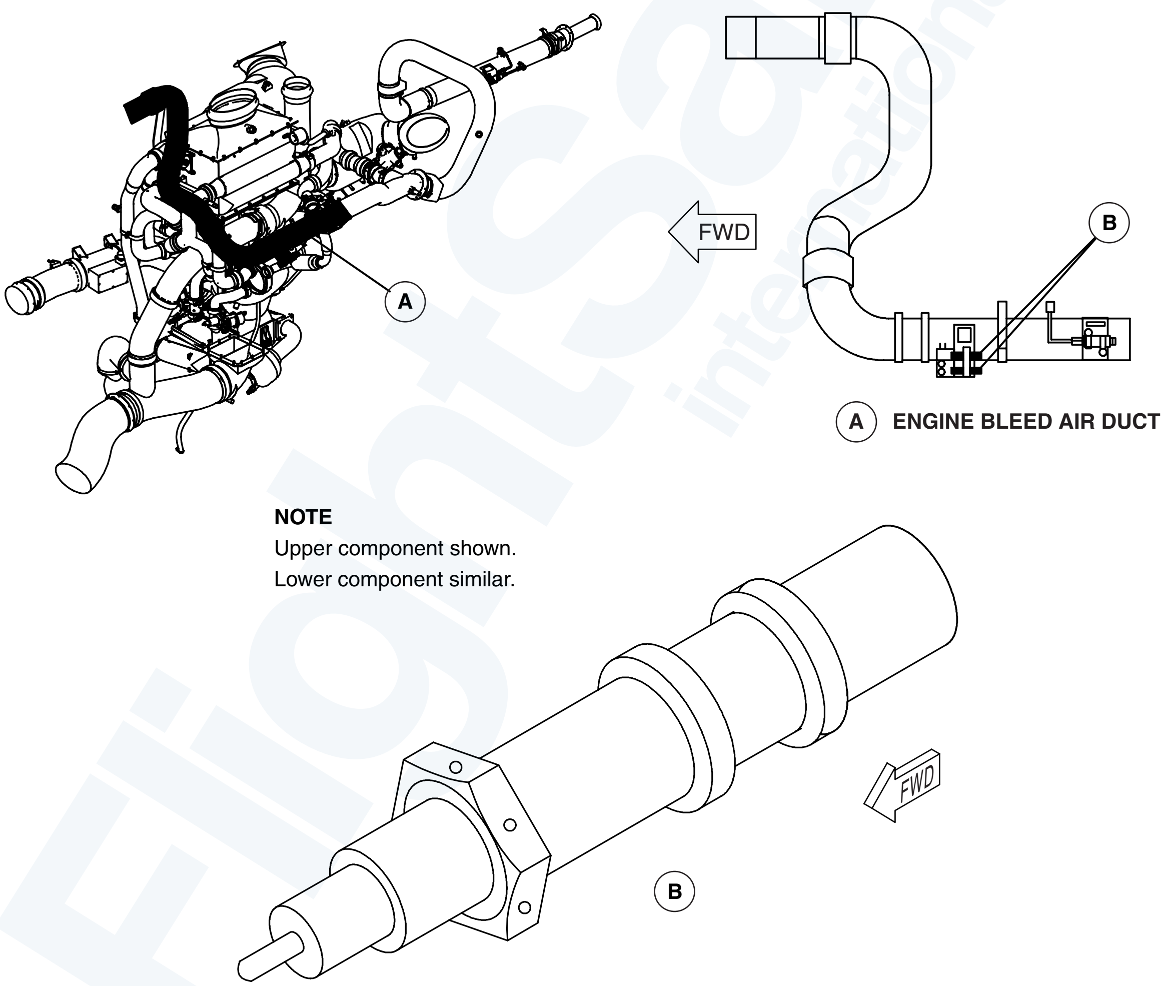

Figure 36-26. Aft Fuselage Duct Leak Overtemperature Switch

APU Bleed Air Ducts

Refer to Figure 36-27. APU Bleed Air Ducts.

Shrouded titanium APU bleed air ducts, installed aft of the rear pressure bulkhead, run forward through the APU firewall and join the aft fuselage duct upstream of the PFCSOV.

APU Check Valve

Refer to Figure 36-27. APU Bleed Air Ducts.

The dual flapper, spring loaded APU check valve allows APU bleed air to flow to the packs. The valve prevents reverse flow back to the APU.

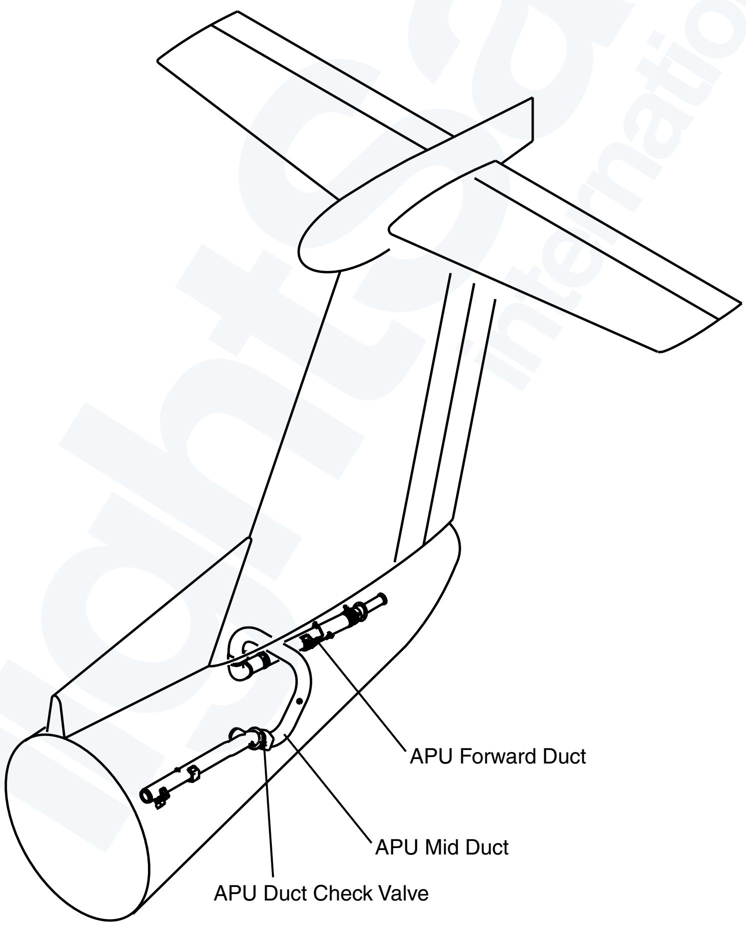

Figure 36-27. APU Bleed Air Ducts

Environmental Control System Electronic Control Unit (ECS ECU)

Refer to:

- Figure 36-28. ECS Electronic Control Unit (ECU).
- Figure 36-29. ECS Interfaces.

The ECU, installed under the center floor boards, responds to pilot switch and sensor commands to conthe BAS sources. The ECU and the BAS have redundant configurations to permit continued ECS operation with mechanical and electrical component malfunctions.

The ECU has two channels designated left and right. Each channel has a digital channel for automatic control and an analog channel for manual control. The analog channel also provides limited backup control should the digital channel fail.

The ECU controls and monitors the bleed air system for the following indications:

- Bleed flow
- Bleed pressure
- Bleed temperature
- Engine bleed supply
- Bleed duct leak detection
- Valve positions.

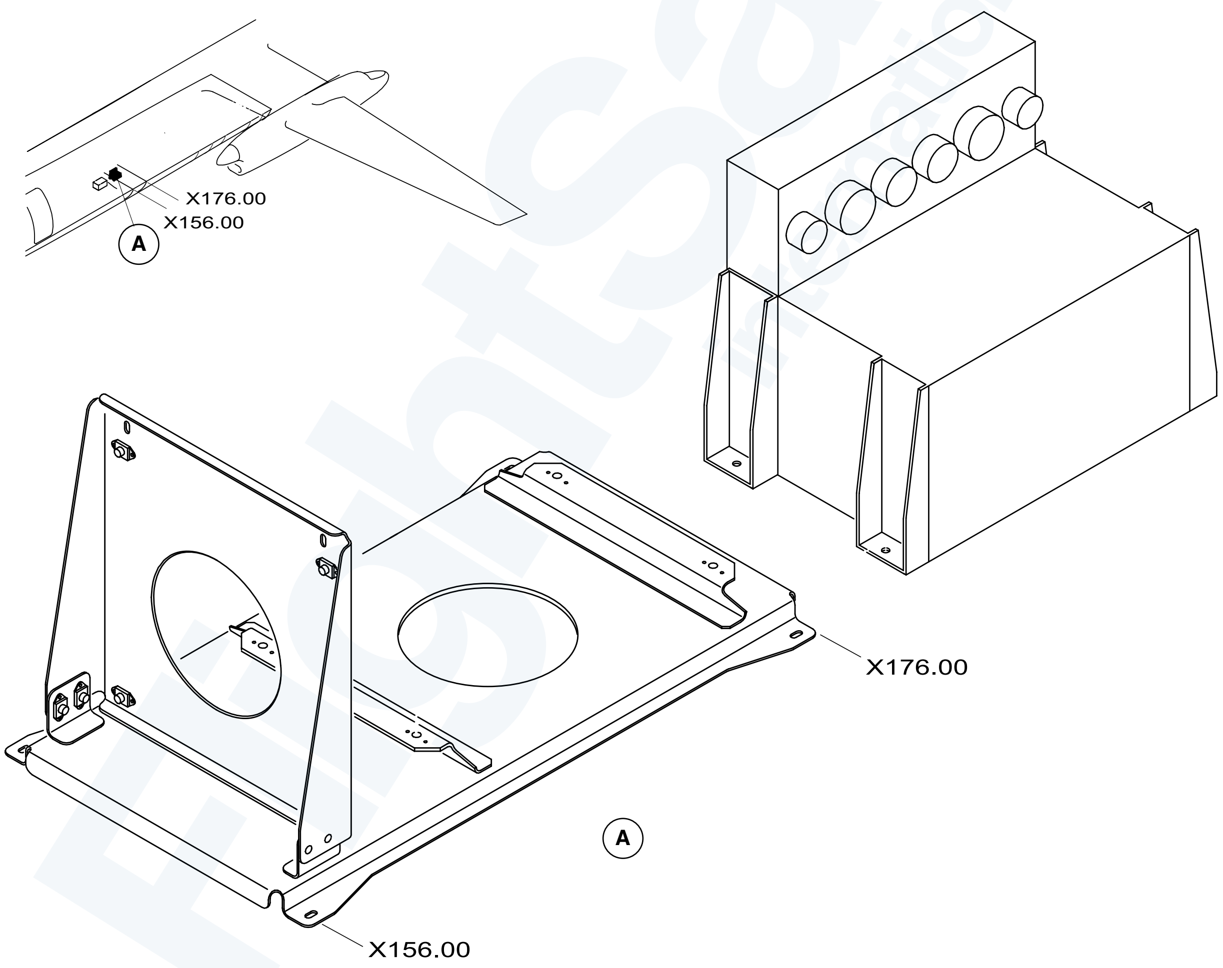

Figure 36-28. ECS Electronic Control Unit (ECU)

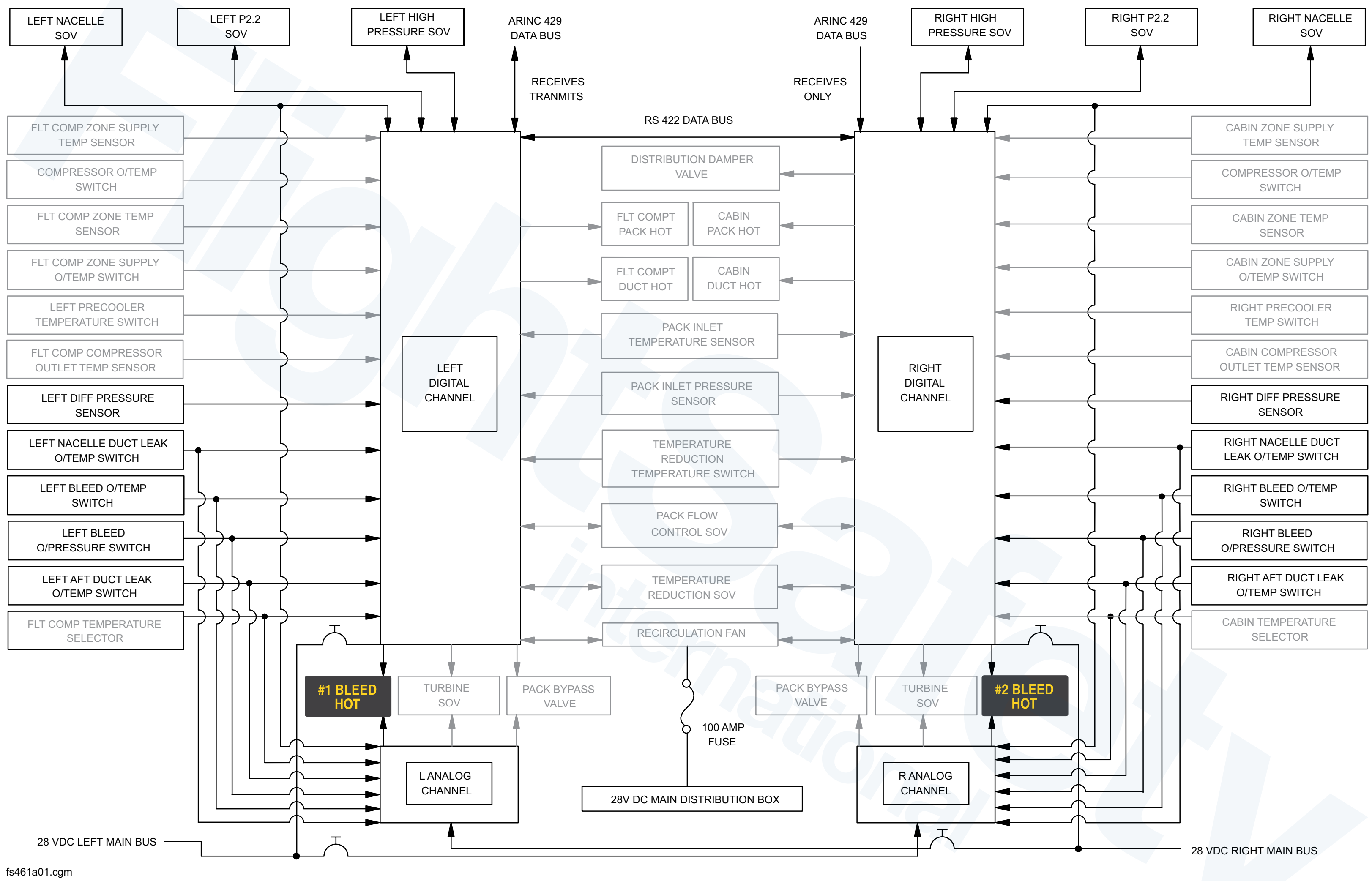

Figure 36-29. ECS Interfaces

CONTROLS AND INDICATIONS

Refer to Figure 36-30. Bleed Air, Air Conditioning Control Panel.

Bleed Control Switches

The bleed air switches are installed in the flight compartment on the air conditioning control panel.

BLEED 1 and BLEED 2 Switches

Selection to ON signals the ECU to open and control NFCSOV.

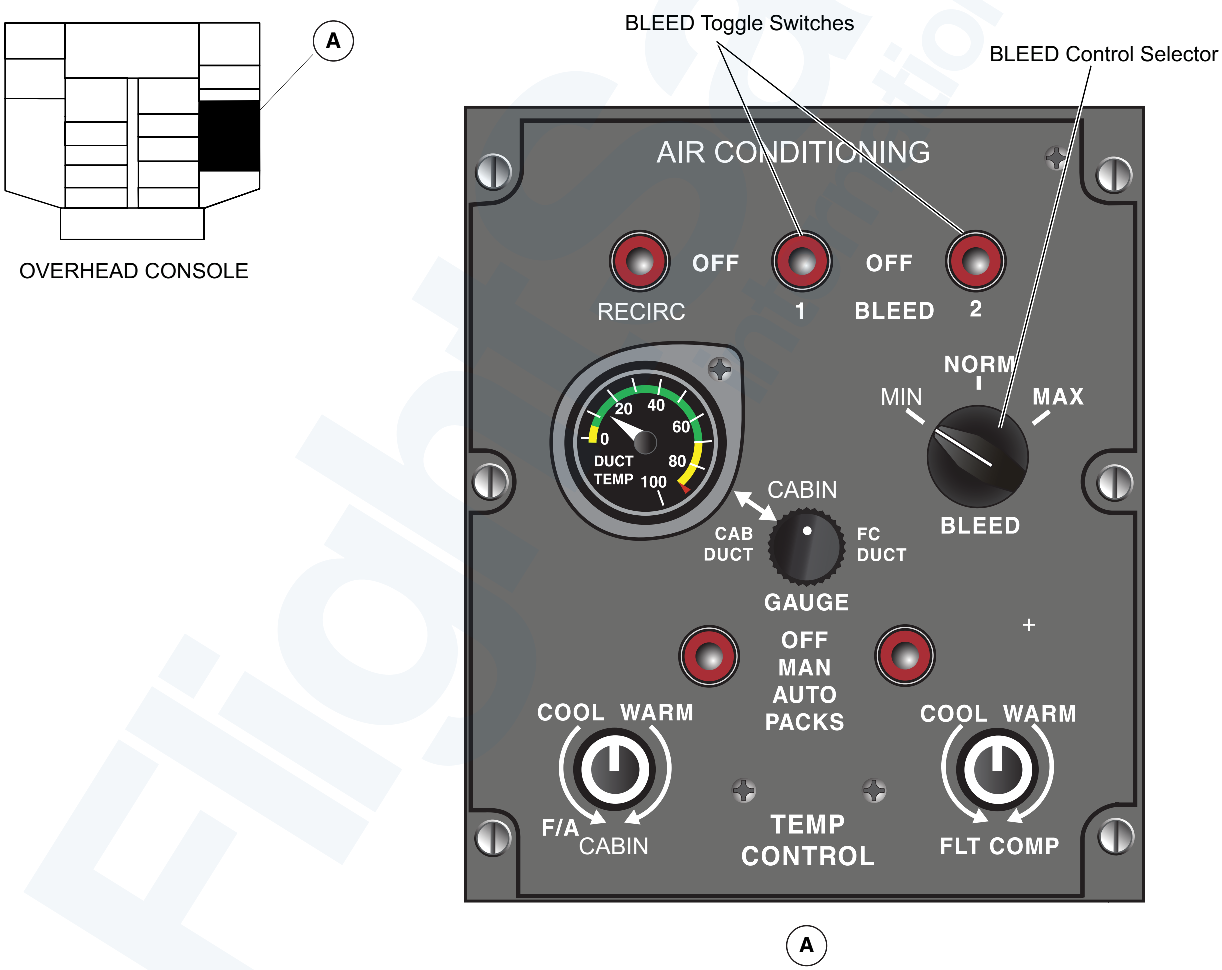

Figure 36-30. Bleed Air, Air Conditioning Control Panel

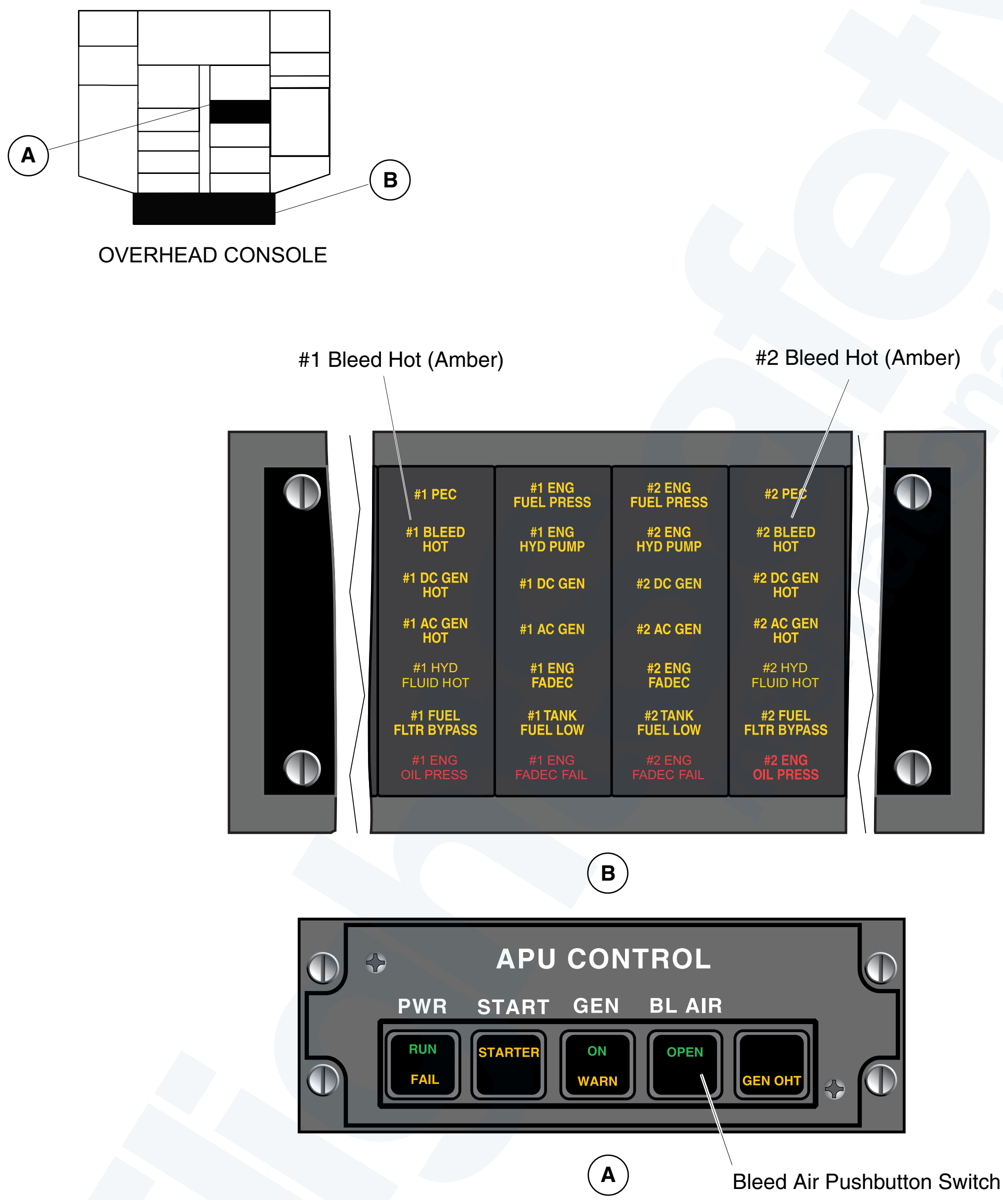

Figure 36-31. Bleed Air System Control and Indication

Bleed Control Selector

Refer to Figure 36-31. Bleed Air System Control and Indication.

The BLEED control selector is a three-position rotary selector that is used to control the amount of bleed air flow from the two engines.

MIN: Causes the ECU to modulate the nacelle flow control shut-off valve to minimum air flow.

NORM: Causes the ECU to modulate the nacelle flow control shut-off valve to normal air flow.

MAX: Causes the ECU to modulate the nacelle flow control shut-off valve to maximum air flow.

The Bleed Air System has the following caution lights:

- No.1 BLEED HOT
- No.2 BLEED HOT.

Refer to Figure 36-32. Bleed Flow Versus Selection and Rating Chart.

The No.1 BLEED HOT caution light illuminates when:

- Left bleed overpressure switch detects a pressure of greater than 100 psig.
- Left bleed overtemperature switch detects a temperature of greater than 349°C (660°F).
- Left nacelle duct leak overtemperature switch detects a temperature of greater than 182°C (360°F).

The No.2 BLEED HOT caution light illuminates on when:

- Right bleed overpressure switch detects a pressure of greater than 100 psig.
- Right bleed overtemperature switch detects a temperature of greater than 349°C (660°F).
- Right nacelle duct leak overtemperature switch detects a temperature of greater than 182°C (360°F).

Both No.1 and No.2 BLEED HOT caution lights illuminate when:

- Both aft duct leak switches detect a shroud temperature greater than 180°C.

NOTE

A BLEED HOT caution light will remain ON until the cause is removed and the respective BLEED switches and/or the APU BL AIR switchlight are cycled through OFF and back to the ON position.

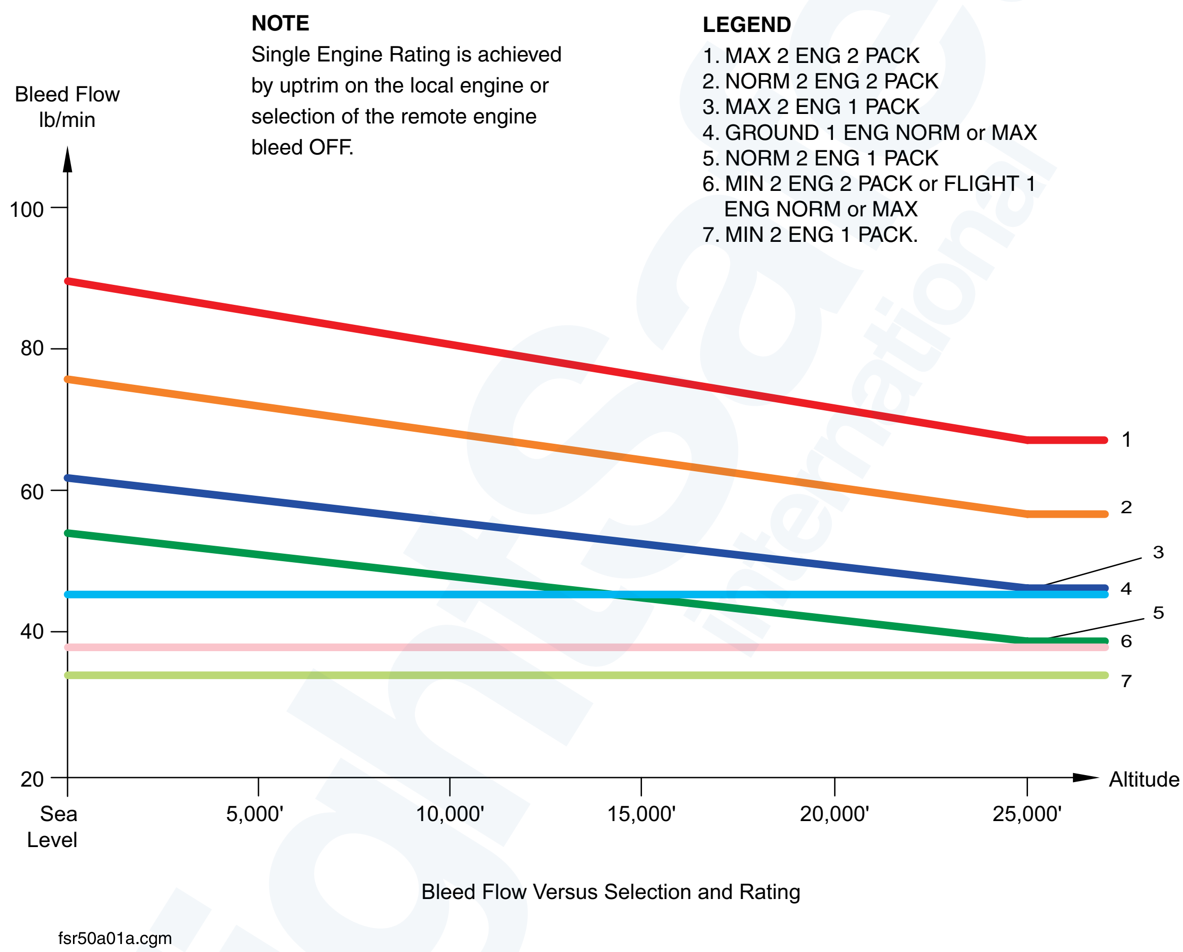

Figure 36-32. Bleed Flow Versus Selection and Rating Chart

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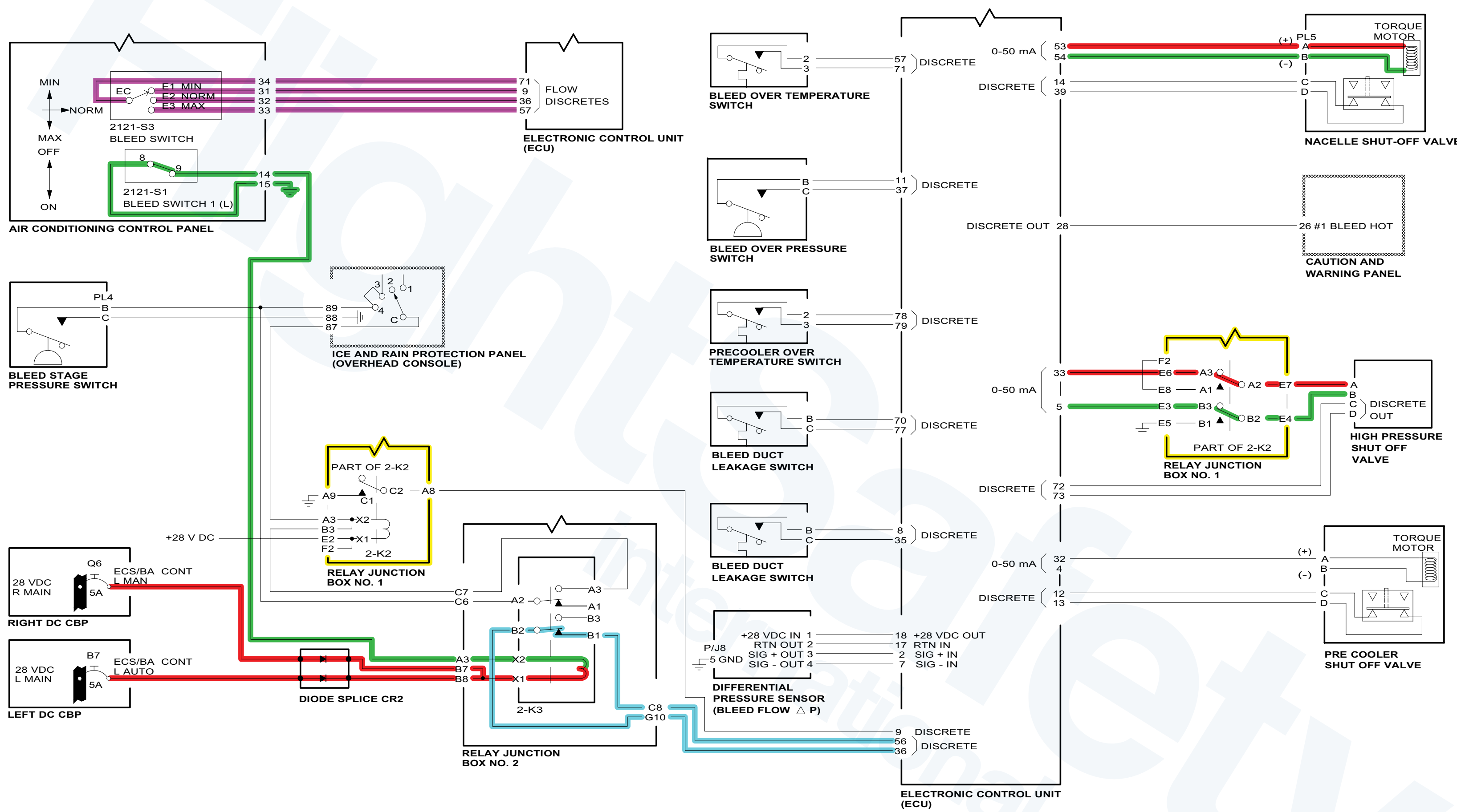

Figure 36-33. Bleed On De-Ice Off

OPERATION

Refer to Figure 36-33. Bleed On De-Ice Off.

NOTE

Left side shown right side similar.

Engine Bleed Air

Selection of bleed switch 1 provides a ground to close relay 2-K3 (Relay JB2). 28 VDC for the relay is supplied from the left and the right main DC bus.

A "Bleed Selected On" discrete signal from the ECU (Pin 36) is returned to the ECU (Pin 56) through relay contacts B1 and B2.

The ECU outputs a signal to open the NFCSOV (Pins 53 and 54).

Excitation voltage from the ECU (Pin 71) is sent to the rotary BLEED (MIN/NORM/MAX) switch and returned through pins 9, 36 or 57.

This input along with differential pressure sensor input is used to adjust the current delivered to the NFCSOV for modulation.

HPSOV torque motor control is through the normally closed contacts (A2 and B2) of relay 2-K2. Valve position feedback is provided through Pins 72 and 73.

NOTES

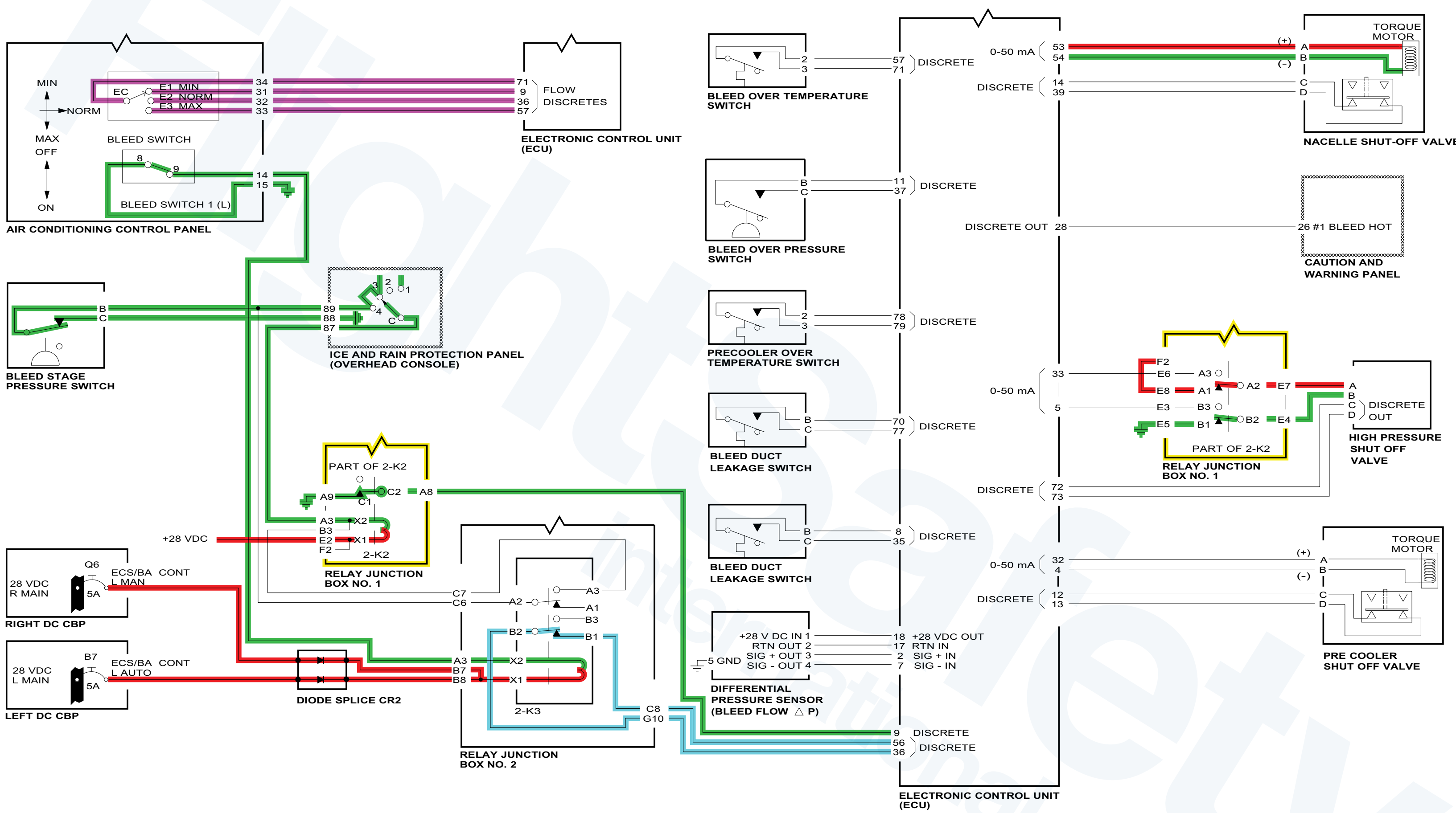

Figure 36-34. Bleed On De-Ice On

BLEED OFF OR DE-ICE ON – HPSOV Control

If the bleed switch is selected off *or* the airframe de-ice is selected on, control of the HPSOV is via the bleed stage pressure switch.

Refer to Figure 36-34. Bleed On De-Ice On.

Relay 2-K3 is energized and relay 2-K2 is relaxed. When the P3 Bleed stage pressure switch senses pressure less than 77 psi (531 kPa), it closes, supplying a ground to energize relay 2-K2.

Energizing 2-K2 removes control of the HPSOV from the ECU and contacts A2 and B2 supply 28 VDC and a ground to open the HPSOV.

NOTES

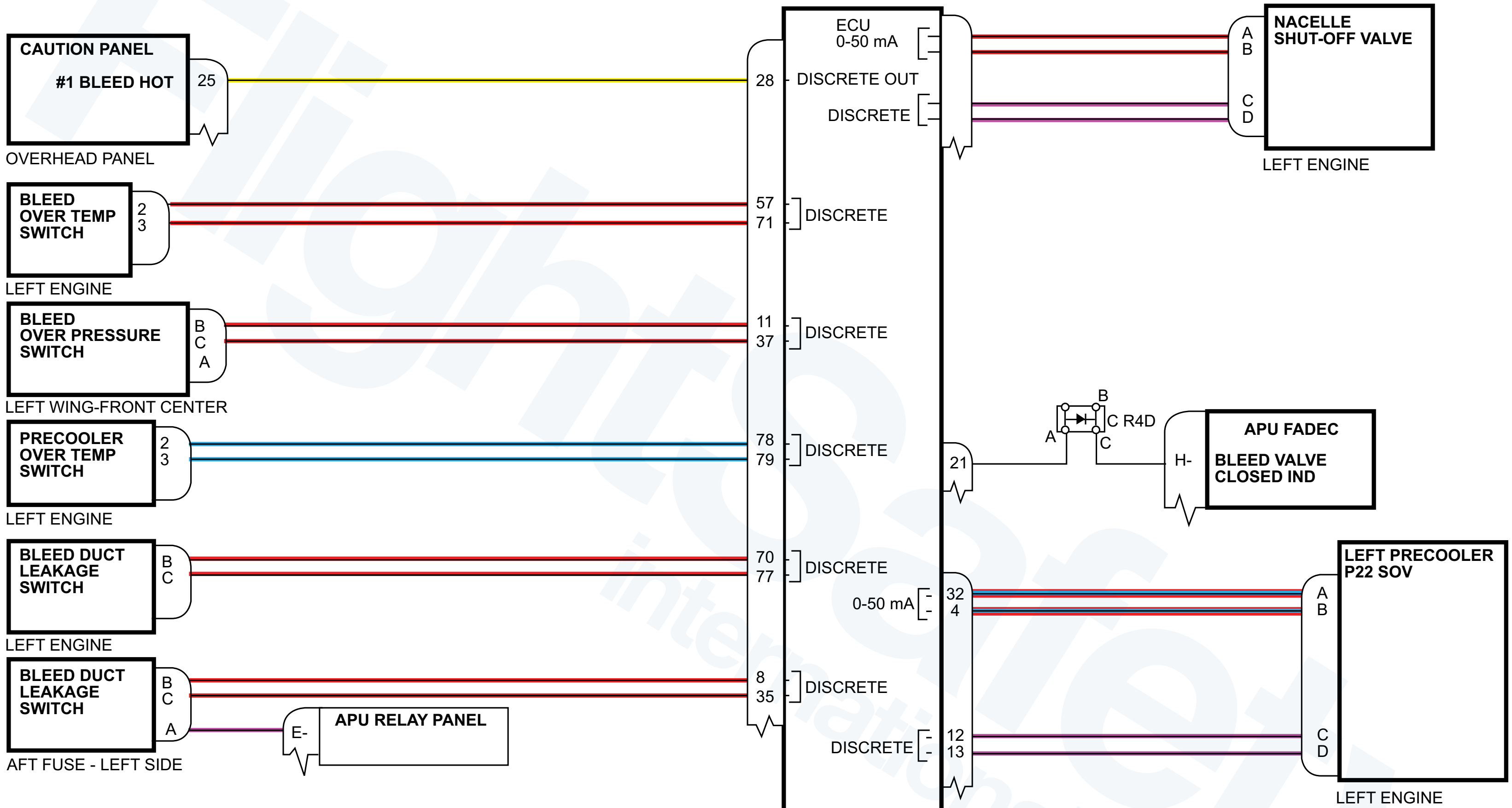

Figure 36-35. Control and Fault

Control and Fault

Refer to Figure 36-35. Control and Fault.

When P3 air temperature is greater than 288ºC (550ºF), the pre-cooler overtemperature switch closes and sends a discrete signal to the ECU (Pin 79). The ECU will command the P2.2 SOV open.

Fault

Bleed overtemperature switch senses temperature exceeding 349°C (660°F), a discrete signal is sent to the ECU Pin 71.

Bleed overpressure switch senses a pressure in excess of 100 psig (689 kPa), a discrete signal is sent to the ECS ECU Pin 37.

Duct leak switch senses temperature in the shroud greater than 182ºC (360ºF), a discrete signal is sent to the ECS ECU Pin 77.

Upon receipt of any of the above discretes, the ECU will shut down the system by sending signals to close the HPSOV, NFCSOV and the P2.2 SOV. The ECU will output a signal to illuminate the associated BLEED HOT caution light and log a related fault code.

The system will not automatically reset. The bleed switches must be selected to OFF then ON to reset the system.

If a leak is detected in the common shroud area (aft fuselage area) and both overtemperature switches sensed the temperature exceed 182ºC (360ºF), a discrete signal is sent to the ECSECU and the APU relay pin E-.

The ECU will send a signal to the APU FADEC for APU bleed valve control. The APU relay panel will interrupt the bleed request and the APU bleed valve will close.

The ECU will output a signal to illuminate both BLEED HOT caution lights. The system will not automatically reset. All bleed switches (Engine and APU) must be selected to OFF then ON to reset the system.

APU Bleed Air

Selection of the BL AIR switch on the APU CONTROL panel energizes the APU bleed air system. The APU will supply the ECS with bleed air only when the two engine BLEED 1 and 2 switches are selected off.

Selection of the BLEED 1 and/or BLEED 2 switches send a signal to the APU FADEC to close the APU bleed valve.

Deice and Oil Cooler Ejector Bleed Air

Bleed air from each engine is always available for the deice system and the oil cooler ejectors.

Engine bleed air flows through a smaller deice pre-cooler in the precooler reducing the deice bleed air temperature. This deice precooler uses a continuous flow of P2.2 bleed air to reduce the temperature of the bleed air before it flows to the deice system.

The oil cooler ejector shutoff valve is opened by the FADEC when the conditions that follow occur:

- Aircraft is WOW
- PLA is less than 60 degrees
- Oil temperature is greater than 104°C.

NOTE

Check valves in the wing ducts prevent the use of APU bleed air for the deice system and oil cooler ejectors.

Centralized Diagnostic System (CDS)

Refer to:

- Figure 36-36. ARCDU, CDS (1 of 2).
- Figure 36-37. ARCDU, CDS (2 of 2).

The CDS lets aircraft maintenance personnel:

- Read malfunction reports from last or previous flight legs
- Read the part number of the software for the left and right channels of the ECS
- Perform fault isolation using the data displayed on pages 6 and 7.

As well, refer to chapter 21 Air Conditioning CDS as both systems share the same ECS ECU.

Crew Report

No.1 BLEED HOT light came on during climb. Bleed No.1 remained in OFF position for the remainder of the flight. What troubleshooting steps will you take?

Figure 36-36. ARCDU, CDS (1 of 2)

Figure 36-37. ARCDU, CDS (2 of 2)

36-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB2111004 Kit - Fluidized Bath
- 500 or equivalent Temperature Bath
- Commercially Available Thermometer (capable of an accuracy of ± 0.9°F (± 0.5°C), or better)
- Keithley 2700 or equivalent Multimeter/Data Acquisition (Temperature and Resistance Measuring) System
- GSB2400001 Multimeter - Hand Held
- Commercially available Air Pressure Regulator (0 - 125 psi) (0 - 861.85 kPa)

36-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 36-11-31-000-801 Removal of the P2.2 Shutoff Valve
- AMM 36-11-31-400-801 Installation of the P2.2 Shutoff Valve
- AMM 36-11-09-160-801 Cleaning the High Pressure Shutoff Valve Filter
- AMM 36-11-19-160-801 Cleaning of the Nacelle Shutoff Valve Filter
- AMM 36-11-31-160-801 Cleaning of the P2.2 Shutoff Valve Filter
- AMM 45-00-21-742-801 Retrieval of Data from the Central Diagnostic System (CDS) − Environmental Control System (ECS)
- AMM 36-11-00-710-801 Operational Check of the Bleed Air Nacelle Shut−Off Valve (this task incorporates 215200–204) MRB#361100–203)
- AMM 36-11-00-710-802 Operational Test of the Bleed Air System
- AMM 36-11-00-720-803 Functional Check of the Wing Bleed Leak Detection Switches (MRB#361100−209)
- AMM 36-11-16-720-801 Functional Check of the Bleed Air Over Temperature Switch (CMR#361100−107)
- AMM 36-11-16-720-802 Functional Check of the Resistance Temperature Device (RTD) Sensor (CMR#361100−110)
- AMM 36-11-36-720-801 Functional Check of the Bleed Air Overpressure Switch (CMR#361100−106)